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Computer Science and Engineering
Software Engineering II

Requirements analysis and specification document
Students & Companies

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1 | Introduction

1.1. Purpose

The "Students & Companies (S&C)" platform addresses the real-life challenges faced by university students in finding suitable internships and by companies in identifying qualified candidates. It aims to simplify the internship matching process by leveraging student skills, experiences, and preferences alongside company requirements and opportunities. By offering features such as personalized recommendations and tools for managing the selection and feedback process, the platform seeks to improve the efficiency, transparency, and overall satisfaction of both students and companies.

The purpose of this RASD is to provide a blueprint for understanding the problem domain, defining the requirements of the platform, and highlighting the functionalities and constraints needed to ensure the successful implementation of S& C. It covers scenarios, domain elements, use cases, and more instruments to guide the development towards a solution that effectively addresses these challenges.

1.1.1. Goals

G1 Students can find suitable internships that match their skills, experiences, and preferences.

G2 Companies can find candidates that meet their internship requirements.

G3 Students and companies can be recommended to one another based on their preferences, profile, and feedback.

G4 Students and companies can monitor and manage the selection processes they're involved in.

G5 Students, companies, and universities can monitor and track the status of internships, complaints, and the progress of candidates.

1.2. Scope

In order to tackle the aforementioned challenges, the platform aims to provide a simple and smooth way for students and companies to find internship opportunities and candidates that match their needs. To achieve this, the platform allows students and companies to create profiles with which they can search for and contact the other party. Students and companies can easily search on the platform for these opportunities or can be given recommendations of possible opportunities that fit their needs.

The platform not only allows both parties to find interesting opportunities, but also offers the possibility of managing the selection process on it. The platform gives tools to both parties to manage as they like the selection, allowing the possibility of creating and answering questionnaires, scheduling and carrying out interviews, and withdrawing, accepting or rejecting.

According to the result of the selection process, the platform also gives the possibility of keeping track of an ongoing internship, not only to each of the parties involved, but also to universities that are registered and have students related to them. During the internship, students and companies can report complaints about their counterpart, which will be received by the university in case they are registered.

Finally, by the end of an internship, both parties will have to send feedback about the experience and their counterpart, feedback which will be used later by the platform when generating recommendations to future students and companies.

In general, the platform offers students and companies the opportunity to not only find suitable opportunities, but also to manage the selection process and carry the internships out.

1.2.1. World Phenomena

- W1 Students have CVs that describe their experiences, skills, and attitudes.
- W2 Companies define projects and terms offered for internships.
- W3 Companies determine availability and preferred dates for interviews.
- W4 A student starts an internship in a company.
- W5 Students and companies conduct an interview outside the platform.
- W6 A student opts out of an internship or a selection process outside the system (e.g., communicates directly with the company).
- W7 Feedback and opinions about internships and students are provided verbally or via channels outside the platform.
- W8 Complaints are made to universities via channels unrelated to the system.

1.2.2. Shared Phenomena

World-controlled shared phenomena:

- WC1 Students register on the platform, providing personal details, proof of student status, and information about their university.
- WC2 Companies register by uploading information about them and their official documents for validation.

WC3 System administrators review documents and validate accounts.

WC4 Students and companies log in to their accounts using their credentials.

WC5 Students upload CVs to the platform.

WC6 Students fill a professional profile, including academic and work experiences, skills, and other details.

WC7 Companies post and CRUD their projects and the terms of the internships.

WC8 Students input preferences for internships.

WC9 Students browse available internships using keywords and filters.

WC10 Companies review student profiles and CVs through the platform.

WC11 Companies contact students if their profiles match internship needs.

WC12 Students apply for internships and initiate contact with companies through the platform.

WC13 Students and companies start the selection process if both are interested in the internship opportunity.

WC14 Students and companies confirm or reject proposed interview schedules.

WC15 Companies upload or input structured questionnaires into the system.

WC16 Students provide answers to structured questionnaires during or after interviews.

WC17 Companies create templates with criteria to evaluate students during the selection process.

WC18 Students participate in scheduled interviews conducted through the platform.

WC19 A student opts out of the selection process at any time through the platform.

WC20 Students and companies make final decisions on internship acceptance or rejection.

WC21 Companies direct complaints to the university of a student doing an internship with them.

WC22 Students provide information or complaints about ongoing internships to universities.

WC23 Students provide feedback and suggestions about the company and the internship at the end of it.

WC24 Companies provide feedback and suggestions about the student at the end of the internship.

WC25 Universities are validated by providing their official details, including proof of legitimacy.

WC26 System administrators contact universities to verify student details or enrollment status.

WC27 Universities reject all the selection processes ongoing related to an internship offer as they have already selected all the candidates they needed.

Machine-controlled shared phenomena:

MC1 The system recommends internships to students based on their profiles, preferences, and CVs.

MC2 The system verifies login credentials for students and companies.

MC3 The system recommends student CVs to companies based on their posted internships.

MC4 The system notifies students about recommended internships.

MC5 The system notifies companies about recommended candidates.

MC6 The system validates interview schedules, checks for conflicts, and updates calendars for both parties.

MC7 The system notifies students when companies try to initiate contact.

MC8 The system notifies companies when students try to initiate contact for internships.

MC9 The system notifies both parties when the selection process starts.

MC10 The system sends interview notifications.

MC11 The system stores and delivers structured questionnaires to students.

MC12 The system stores templates created by companies for annotating and evaluating candidates.

MC13 The system tracks application statuses (e.g., submitted, accepted, rejected).

MC14 The system finalizes selections by logging the outcome of the company's selection process (e.g., selected, rejected, pending).

MC15 The system finalizes the selection if the student opts out of the process (e.g., starts an internship with another company).

MC16 The system tracks the state of ongoing internships.

MC17 The system stores feedback provided by users.

MC18 The system redirects complaints made by users to universities.

1.3. Definitions, Acronyms, Abbreviations

1.3.1. Definitions

Internship: A temporary position offered by a company to students or recent graduates to gain work experience in a specific field.

Recommendation Engine: A system feature that matches students and companies based on preferences, skills, and requirements.

Keyword Matching: A functionality within the recommendation engine that identifies potential matches using predefined keywords in profiles and internship descriptions.

System Administrator: A user responsible for validating accounts and documents, monitoring platform activity, and ensuring system functionality.

Feedback: Evaluative information provided by students and companies about their experiences with the internships .

Complaint: A formal expression of dissatisfaction or concern submitted by students or companies during or after an internship.

Application : We will refer to the process in which a students applies for an internship offer or a company contacts a student to start a selection process, provided both parties agree, as "Starting contact" or "Initiating contact"."

Rating System : A component of the recommendation engine within the platform that evaluates and ranks internship opportunities and student profiles based on feedback and performance metrics. The rating system aggregates feedback provided by students, companies, and universities to determine the quality and relevance of internships and candidates. It uses weighted scores derived from factors such as user ratings, feedback history, and past interactions to prioritize recommendations and improve the matching process. This ensures that higher-rated internships and candidates are more prominently recommended to users.

Email provider : A third-party service or platform integrated into the system to manage the sending, receiving, and verification of emails. The email provider is responsible for sending account confirmation emails, password recovery links, and notifications regarding platform activities (e.g., new internship applications, contact requests, or interview reminders).

1.3.2. Acronyms

API: Application Programming Interface. A set of functions and procedures that allow different software components to communicate with each other.

CV: Curriculum Vitae. A document detailing a person's education, qualifications, and previous experience, typically used for job applications.

S&C: Students & Companies. The name of the platform described in this document.

WCAG: Web Content Accessibility Guidelines, a set of recommendations for making web content accessible to users with disabilities.

GDPR: General Data Protection Regulation. A regulation enacted by the European Union to strengthen data protection and privacy for individuals within the EU.

CRUD: Create, Read, Update, Delete. Refers to the basic operations performed on data or resources in a database.

1.3.3. Abbreviations

WC* : Shared phenomena World-controlled.

W* : World phenomena.

MC*: Shared phenomena Machine-Controlled.

UC* : Use Cases

G* : Goal

R* : Functional requirement

D* :Domain assumption

1.4. Revision History

- Version 1.0 22/12/2024

1.5. Reference Documents

The document is based on the following materials:

- The specification of the RASD assignment of the Software Engineering II course a.a. 2024/25
- Slides of the course on WeBeep

1.6. Document Structure

- **Introduction:** This section provides an overview of the project, emphasizing its purpose, objectives, and the motivations behind its development.
- **Overall Description:** This section offers a high-level overview of the system, detailing the interactions between the world, the machine, and their shared phenomena. It also includes the domain description and outlines the key assumptions considered during development.
- **Specific Requirements:** This section presents a comprehensive analysis of the system's requirements, including functional and non-functional aspects, along with hardware and software constraints necessary for implementation.
- **Formal Analysis:** This section provides a formal representation of the world phenomena using the Alloy language, offering a precise and rigorous description of system behaviors.
- **Effort Spent:** This section summarizes the effort dedicated to creating this document, detailing the time allocated to each section and task.

- **References:** This section lists all the sources, references, and materials consulted during the preparation of this document, ensuring proper acknowledgment of external contributions.

2 | Overall Description

2.1. Product Perspective

2.1.1. Scenarios

Company registration

The company "Tech" decides to register on the platform to publish internship opportunities. A representative logs into the platform and selects "Register as Company." The representative provides email, password, and then the system sends an email verification to them with a link to confirm the email address. Once the email is verified, the company can input information about them and Official Documents (Business registration certificate). The company's status is set to "In Review", where they can log into the system but cannot use all the functionalities it provides. A system administrator reviews the documents. The administrator accepts the registration, and the company's status updates to "Approved". The system sends a notification, allowing the company to log in and complete their profile.

Student registration

A student named "Alex" visits the platform and selects "Register as Student." Alex fills in the registration form with their name, university email and their password. The system sends a confirmation email with a verification link. Alex verifies the email, and their account is activated. Then, he can input his personal data and professional profile, as well as upload his CV and provide proof of his status as a student. He only sends documents that link him with his university, fills out his professional profile and inputs some of his personal data, as is the minimum required. A system administrator reviews the documents. After, Alex logs in and uploads their CV in PDF format. Alex also updates their profile with additional details, including preferred industries, skills, and languages. The profile status changes to "Approved" after the system administrator validates his documents, allowing Alex to start browsing internships.

University registration

A university called “The MCA” is interested in registering to the platform, as it has noticed that many of its students are using the platform to seek out internship opportunities. As such, they go to the page and select “Register as University”. They input the associated email and password and confirm such email. After, they are asked to fill in the registration form with the name of the university, some information about it, and proof of its official status as an institution. After some time, the system administrators review and approve the documents, setting the status of the university account to “Approved”, meaning that now any of its students that want to register to the platform can be linked with the university directly.

Student updates their profile

“John” logs onto the platform and goes to his “Profile” section. Here, he wants to update his profile, since during the registration he inputted his professional profile and his personal data but never uploaded his CV. Now, in the section, he clicks on “Update profile”, which loads the form to update the data related to his profile. Now, he uploads his CV in a valid format, sets his preferences for internship offers and saves changes. The system displays a success message, and his profile section is reloaded.

Internship publication by a company

“Tech” logs into the platform and selects "Create Internship". The representative fills in the form with the details: Title: "Data Science Intern," Role Description: "Work on machine learning models and data pipelines," Duration: 6 months, Location: Remote, Remuneration: €1000/month, and Requirements: Python, knowledge of TensorFlow, and a bachelor’s degree in computer science. The representative clicks on “Post internship”. The system saves and posts the internship. Students browsing the platform can now see and apply to it.

Student applies for an internship

Alex finds the "Data Science Intern" position using the platform’s search filters, such as "Remote" and "Python”. Alex reviews the internship details and clicks "Start contact". After submission, the system notifies Tech about the new application.

Internship matching system

The system evaluates Alex’s profile and preferences. Based on the internship database, it recommends relevant internships, such as "Data Science Intern" by Tech and "Machine Learning Assistant" by Innovations. Alex receives a notification about these recommendations. Alex reviews the options and decides to apply to the Tech internship.

Selection process for an internship

Tech reviews Alex’s application and agrees to the contact, so it starts the selection process. The company offers three interview slots: June 1st, 10:00 AM; June 2nd, 3:00 PM; and June 3rd, 1:00 PM. Alex selects the June 2nd slot. The system updates both calendars and sends reminders. On the interview day, Alex joins the video call integrated into the platform. During the interview, the company representative is writing down details and annotations about Alex’s profile in a template created beforehand. Post-interview, Tech evaluates Alex using a structured questionnaire hosted on the platform. The company finalizes the selection and sends an offer to Alex.

A Company receives recommendations for student profiles

The company "Green" publishes an internship for an "environment Analyst", requiring skills such as data analysis and proficiency in Excel. The system analyzes the posted internship's criteria and identifies three student profiles matching Green's requirements. Green receives a notification with a list of recommended candidates: Alice Johnson (Data Science, Environmental Studies, proficient in Excel), Mark Taylor (Sustainability Management, Advanced Excel), and Sarah Brown (Environmental Engineering, experience in data visualization). The company reviews the recommended profiles and selects Alice Johnson to start the contact.

A Student receives internship suggestions

Anna, a student in Computer Science, logs into S & C and completes her profile, specifying her interests in AI and web development. The system processes Anna's preferences and searches for relevant internships. Based on Anna's skills and profile, the system identifies suitable internships, such as "AI Research Intern" at VisionTech and "Frontend Developer Intern" at WebWorks. Anna receives a notification about these opportunities and decides to apply for the "AI Research Intern" position after which VisionTech is notified of her interest.

A student creates a complaint about an internship

John, a student completing an internship at "Global Solutions," encounters a problem: John logs into S & C and navigates to the "Complaints" section. He fills out the complaint form, providing details about the issue and describing his challenges. A notification confirms that John's complaint has been submitted successfully.

A university monitors the progress of its students

The university administrator, Dr. Emily Carter, logs into S&C to monitor the progress of students from her institution which displays active internships and completed internships. Dr. Carter clicks on an individual student's record to review their internship status, including the student's name, the internship's title, and their progress.

A student withdraws from an internship application

Sarah, a student who applied for multiple internships, receives an offer from her top choice and decides to withdraw from other applications. She logs into S&C, navigates to "My Applications," and selects the internship at "BlueTech." She clicks "Withdraw Application" and provides a reason: "Accepted another offer." The system notifies BlueTech about Sarah's withdrawal, and her application status updates to "Withdrawn".

A company selects a final candidate

BrightTech receives several applications for their "UX Designer Intern" role. The HR manager logs onto the platform to manage the selection process. They use S&C's tools to review applications, shortlist five candidates, schedule interviews, and set up a structured questionnaire for evaluating UX skills. As candidates complete the interview process, the HR manager reviews their responses and selects the top candidate for the internship. The system notifies the selected candidate and updates their selection process status to "Selected".

Tracking internship completion

Alex completes his internship at VisionTech. The company updates his internship status to "Completed" and provides final feedback. Alex logs onto the platform and confirms the completion of his internship. Both parties rate their experience and provide feedback, and the data is stored for future analysis.

2.1.2. Domain class diagram

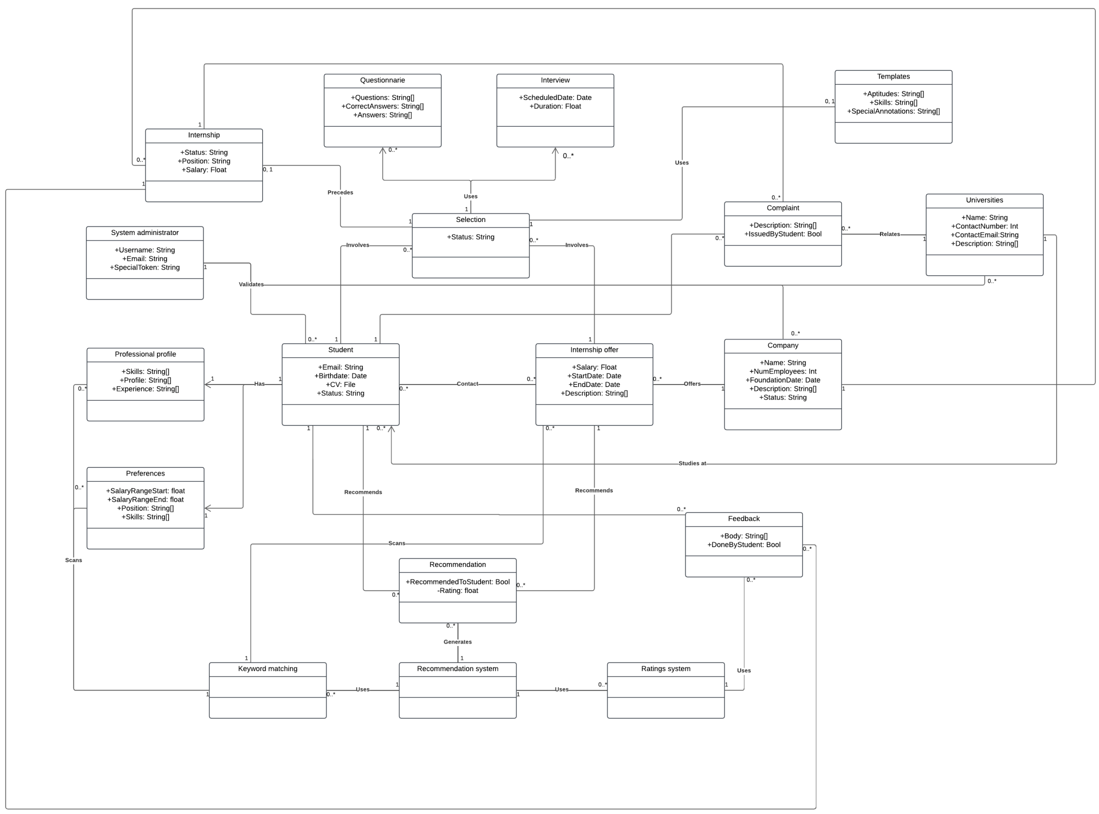

Figure 2.1: Domain diagram for the S&C platform.

In the presented figure, the domain class diagram illustrates all the elements within the domain and their interactions. At a high level, students are required to complete a professional profile, set their preferences, and upload a CV. They can then apply to internship offers by initiating contact with a company. Similarly, companies can reach out to students if they find their profiles suitable for an internship position.

This interaction can happen naturally as students and companies browse the platform or through recommendations provided by the system. The recommendation system utilizes two specialized mechanisms: the keyword matching system and the rating system. These systems analyze students' preferences, professional profiles, and the details of internship offers to generate personalized recommendations.

Once one party initiates contact and the other accepts, the selection process begins. During this phase, both students and companies can leverage tools offered by the platform, such as creating questionnaires or scheduling interviews, to manage the process. If the company determines that a candidate is suitable, they can select the student for the internship, and the internship officially starts.

During the internship, both students and companies have the option to file complaints. If the student's university is registered on the platform, these complaints are forwarded to the university for review. Universities can also monitor the status of their students' internships and review any complaints related to them or the company.

At the conclusion of the internship, both students and companies are prompted to provide feedback and rate their experiences with their counterpart. This feedback is stored for future use by the platform to improve recommendations and user satisfaction.

Additionally, the system administrator plays a crucial role in the process by validating accounts for students, companies, and universities. This validation involves reviewing the official documents submitted by each user to ensure their legitimacy and compliance with platform requirements.

2.1.3. State diagrams

The state diagrams here presented serve to describe the states of some of the most important processes that occur in the system, and how they evolve as the process goes on.

Registration process

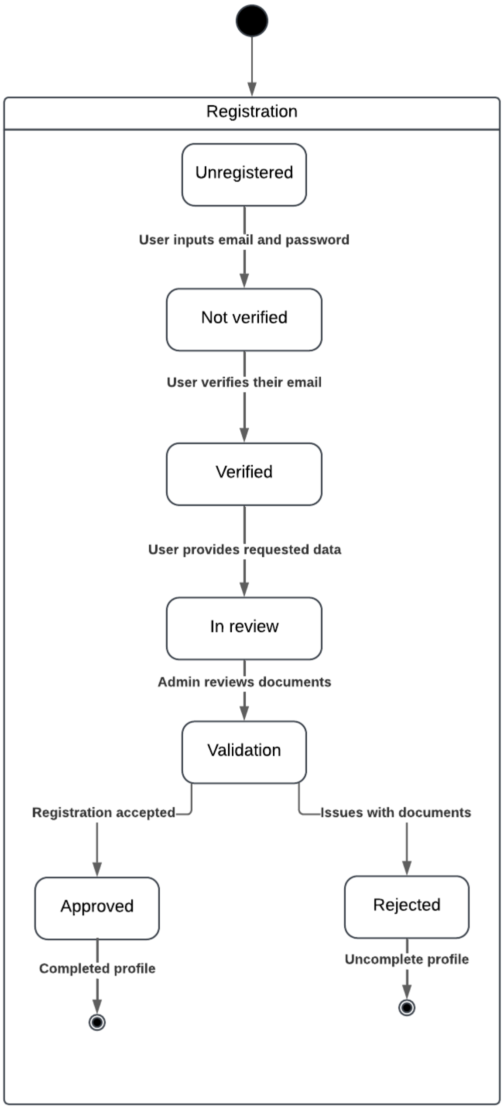

Figure 2.2: Registration process state diagram

During the registration, the initial state of the user is clearly unregistered. After they input their email and password, they go on to be an unverified account, as they have to confirm their email address by confirming the mail that was sent by the system. After confirming, they become verified and need to send their official documents for verification. After sending their documents, their state updates to in review. Then, the system administrator has to check the documents and update the state of the account to either approved, which means that the registration has been completed, and the user has full access to the functions of the system, or to rejected, which means that the profile is incomplete and must send proper documentation.

Selection process

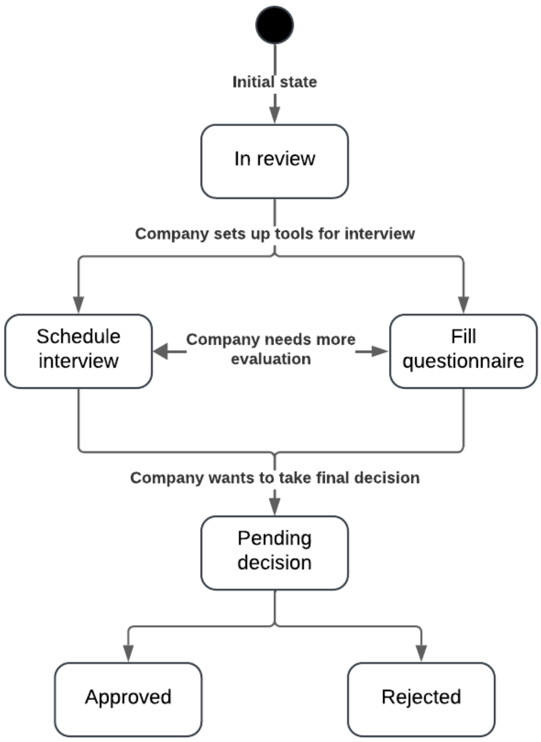

Figure 2.3: Selection process state diagram

Initially, the selection process is marked as "in review," indicating that the company is still evaluating the student. This serves as the default state for the entire process, remaining unchanged while the student completes questionnaires and attends interviews arranged by the company. Once the company makes a final decision regarding the student, the selection process transitions to the appropriate state, either "approved" or "rejected."

Ongoing internship - Complaint

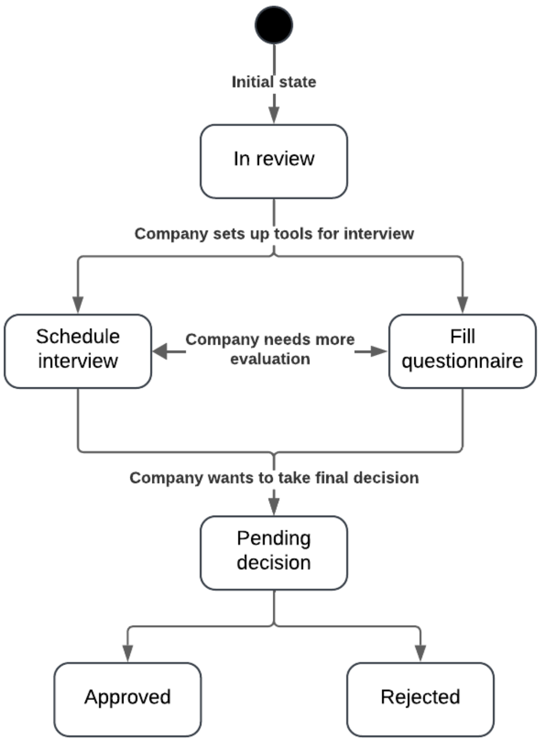

Figure 2.4: Internship state diagram

The complaint state diagram gives a high view of the entire internship process. At the beginning, the internship starts, and as progress is being assessed, the company or the student can decide whether to report a complaint. If the complaint is made, then the university, in case it's registered in the system, shall be notified of it. In either case, whether or not the complaint is reported, the internship shall continue until its end date, and by the end of it both parties must leave their feedback about their counterpart.

Recommendation system

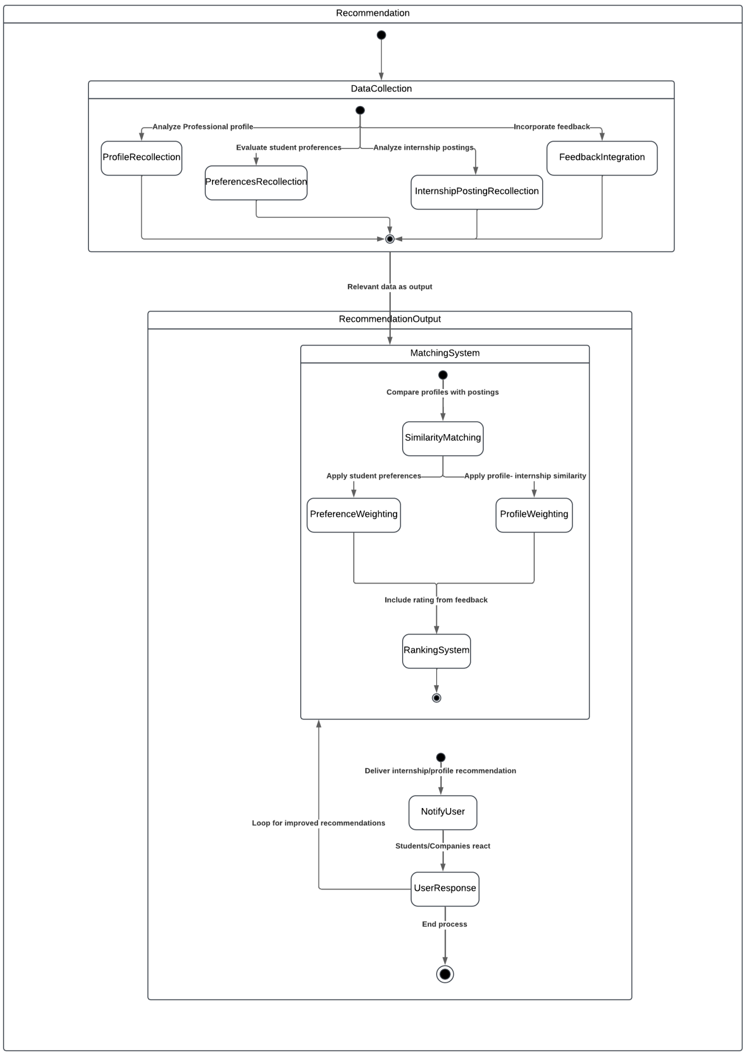

Figure 2.5: Recommendation state diagram

The recommendation system is a complex process involving multiple states and steps. It begins with a stage of data collection, gathering information such as preferences, professional profiles, internship postings, and feedback. Once this data is collected, the system performs a matching process, aligning similar student profiles with the needs of internship offers.

For student recommendations, the system weighs their preferences and the content of the internship postings to identify the best-fitting opportunities. Conversely, for company recommendations, it evaluates the professional profiles of students alongside the company's requirements to identify the most suitable candidates.

Feedback provided by other students and companies is then utilized to rate both the companies and candidates, which, in turn, enhances the recommendations associated with them. Users are presented with a selection of the top recommendations tailored to their needs. Based on the actions users take regarding these recommendations, the system can generate a new set of suggestions if the initial options do not meet their expectations.

2.2. Product functions

Registration process

The registration and login functions are available to all expected users of the system (students, companies, and universities). Users are required to input their account information, including but not limited to their email and password. After providing this information, they must verify their email by clicking the confirmation link sent by the system.

For students, additional steps include providing personal information, completing their professional profile, and uploading proof of their student status. This proof can be in the form of a certificate of enrollment or an official university card displaying their name and relevant personal details. Students must also supply the name and contact information of their university. If the institution is already linked to the platform, students can simply select it from a list. Uploading a CV is optional during registration and can be completed later.

For companies, registration requires submitting official documents as proof of their legal status. These documents are necessary for account approval and validation. Similarly, universities seeking to be linked to the platform must provide proof of their official status as recognized institutions.

Once the registration process is complete, the user's account is set to a "pending" status. A system administrator reviews the submitted proof documents. Upon validation, the account is activated, granting the user full access to all platform functionalities.

Internship posting

Once their account has been approved, companies can seamlessly post internship offers on the platform. The platform provides a user-friendly template for companies to complete, which includes fields such as the title of the internship, location, salary, duration, expected skills, and a brief description of the responsibilities associated with the position. Additionally, companies can include details about their organization, the working environment, or any other relevant information they wish to share.

Start contact

The application function is available to students and, to a certain extent, companies. Students can browse and review all available internships that interest them. They have the flexibility to apply to as many opportunities as they wish, thereby initiating contact with the respective companies. If a company reviews a student's profile and determines the candidate is suitable, they can accept the contact request, which begins the selection process.

Similarly, companies can initiate contact with students whose profiles they find promising, either through the recommendation system or their own search. In such cases, if the student is interested in the company's offer, they can accept the contact request, and the selection process will commence.

Selection process

Once contact is mutually accepted, the selection process begins. This process is primarily managed by companies, allowing them to create questionnaires for candidates to complete, schedule interviews, and conduct them. The platform offers flexibility, enabling these questionnaires and interviews to occur either on the platform or externally, with companies having the option to input a brief summary of the outcomes.

Companies can also utilize templates linked to student profiles to document details such as strengths, weaknesses, and other relevant considerations. On the other hand, students can actively participate in managing their selection process by completing questionnaires within the allotted time or requesting an interview reschedule if they are unable to attend at the proposed time, providing a valid reason.

Ultimately, companies have the ability to select or reject candidates based on their selection criteria, while students retain the option to opt out of the selection process if necessary.

Recommendation system

The recommendation function of the system leverages various criteria to generate tailored suggestions for both students and companies. It employs two key systems: the keyword matching system and the rating system.

The keyword matching system analyzes students' preferences and professional profiles alongside the internship offers posted by companies. This process identifies internship opportunities that align with students' preferences and matches students whose profiles meet the requirements specified in company postings.

Following this, the matching results are ranked using the rating system, which considers feedback from previous students and companies. This ensures that the highest-rated companies and students are prioritized in the recommendations.

Once recommendations are provided, it is up to the students and companies to decide whether to initiate contact with the suggested party.

Complaint process

After the selection process concludes with a student being selected by a company, the internship begins. During the internship, both companies and students have the option to issue complaints if necessary. These complaints are forwarded to universities, enabling them to monitor the situation and determine appropriate actions.

However, if the company is not registered on the platform, complaints must be submitted through external communication channels.

2.3. User Characteristics

We can find mainly 3 types of users that can use the platform and the specific functionalities available to them. They all have access to the log in and register, and after they have been verified by a system administrator, they have access to the main functionalities of the platform.

2.3.1. Student

Students can search for and explore available internships on the platform, applying to those that interest them by initiating contact with a company. They can participate in multiple selection processes for various internship offers and, if selected, proceed with an internship for the specified position.

2.3.2. Company

Companies can post internship offers, making them accessible to students on the platform. They can manage multiple internship offers and oversee corresponding selection processes. These processes can ultimately lead to internships when suitable candidates are selected.

2.3.3. University

Once registered and approved, universities can efficiently monitor the status of internships involving their students. They can also review complaints submitted by or against their students to ensure proper oversight.

2.4. Assumptions, dependencies and constraint

2.4.1. Regulatory policies

When implementing S&C, it is essential to adhere to regulatory policies and industry standards. The platform must obtain user consent for data collection, ensuring transparency and compliance with the General Data Protection Regulation (GDPR). Additionally, S&C should inform users about the ownership and usage rights of CVs, project descriptions, and other content shared on the platform.

Furthermore, the recommendation and matching processes must be entirely bias-free to uphold principles of equity and fairness. The platform's design should also conform to Web Content Accessibility Guidelines (WCAG) to ensure accessibility for users with disabilities.

2.4.2. Domain Assumptions

Domain assumptions are the foundational beliefs or conditions assumed to be true within the operational environment of the system. These assumptions are critical for the design, implementation, and functionality of the platform and must hold for the system to function as expected. Below are the key domain assumptions for the S&C platform:

D1: Universities provide a way for students to demonstrate their enrollment (e.g., certificates, student cards, or online platforms).

D2: System administrators diligently validate registrations and documents submitted by students and companies.

D3: Students upload CVs in valid and readable formats.

D4: Students provide accurate and regularly updated information about their skills, experiences, and preferences in their CVs and professional profiles.

D5: Companies create detailed and truthful internship postings, including project descriptions and terms.

D6: Companies maintain an organized internal process for reviewing and selecting candidates.

D7: Feedback submitted by users is honest and constructive.

3 | Specific Requirements

3.1. External Interface Requirements

3.1.1. User Interfaces

The S&C user interface is designed to be intuitive, responsive, and accessible to students, companies, and universities. It features a clean, modern layout with clear navigation and responsiveness across all devices.

For students, the interface includes a personalized dashboard, a profile builder, internship search with advanced filters, selection process management, internship tracking, and notifications.

For companies, the interface provides a professional and streamlined design, offering tools to post and manage internships, access candidate recommendations, track internships, and utilize templates for saved candidate profiles.

For universities, the platform offers an organized and authoritative interface, enabling them to monitor internships and review related complaints. The layout is structured to support oversight and effective control.

Overall, the user experience (UX) is action-oriented, transparent, and supportive, with clear guidance on platform functionalities to ensure a seamless and efficient experience.

3.1.2. Hardware interface

Since the system will be fully operational on the internet, the only hardware interface needed to access the platform is a device that allows web browsing, such as a computer or a smartphone.

From the system perspective, the platform can be hosted in a cloud or on-premises environment that ensures high availability and high scalability. This environment shall support secure data storage and backup systems.

3.1.3. Software interfaces

The S&C platform relies on the following software interfaces to provide the full functionality of its services:

Calendar API: Integration with a calendar platform is of high need as it provides a way for companies to schedule interviews.

Video conferencing API: The system can possibly feature support for an external service to conduct remote interviews.

Email provider: The use of services that allow sending confirmation mails to the users creating accounts is essential for the registration and validation of new accounts.

3.2. Functional Requirements

Registration and Account Management

R1: The system allows students to register by providing personal details, professional profile, their CV, login credentials, and proof of student status.

R2: The system allows companies to register by providing official documents, company details, and login credentials.

R3: The system allows universities to register by providing official details and proof of legitimacy.

R4: The system allows students to input university contact information in case the university is not registered.

R5: The system allows system administrators to verify and validate students', universities' and companies' official documents.

R6: The system must allow users (students, companies, and universities) to log in with verified credentials.

R7: The system must allow users to view and update their account information.

Posting and Configuring Profiles

R8: The system allows students to upload and update their CVs, including skills, experiences, and preferences.

R9: The system allows students to maintain a professional profile with academic and work experiences, skills, and additional details.

R10: The system should allow companies to create, update, and manage internship postings with detailed descriptions and terms, as well as delete them while keeping the selection processes they have related to that internship post.

R11: The system provides a way for companies to automatically reject all ongoing selection processes once they have selected the number of candidates they desire for an internship offer.

Internship Search and Candidate Matching

R12: The system allows students to search for internships using keywords and filters (e.g., location, skills, duration, type).

R13: The system recommends internships to students based on their profiles, preferences, and feedback.

R14: The system recommends student profiles to companies based on their internship requirements and feedback.

Application and Selection Process

R15: The system must allow students to apply for internships through the platform.

R16: The system must allow companies to initiate contact with students whose profiles match their needs.

R17: The system must notify students and companies when new contact requests are received.

R18: The system must initiate the selection process if both the student and the company agree to contact.

R19: The system should notify users about the start and updates of the selection process.

R20: The system allows companies to schedule interviews with students.

R21: The system validates interview schedules for conflicts and updates both parties' calendars.

R22: The system provides a tool for conducting interviews through the platform.

R23: The system allows companies to create and manage structured questionnaires for evaluating students.

R24: The system provides a template for companies to write annotations and details about candidates and their abilities.

R25: The system must allow companies to finalize internship selections.

R26: The system must allow students to opt out of the selection process at any point.

R27: The system notifies users about the outcome of the selection process.

R28: The system notifies companies if a student opts out of the selection process.

Monitoring and Feedback

R29: The system keeps track of the ongoing state of internships, including milestones and progress.

R30: The system must provide a way for students and companies to direct information and complaints to registered universities.

R31: The system must request feedback and suggestions from students and companies at the end of the internships.

3.2.1. Use Case Diagram

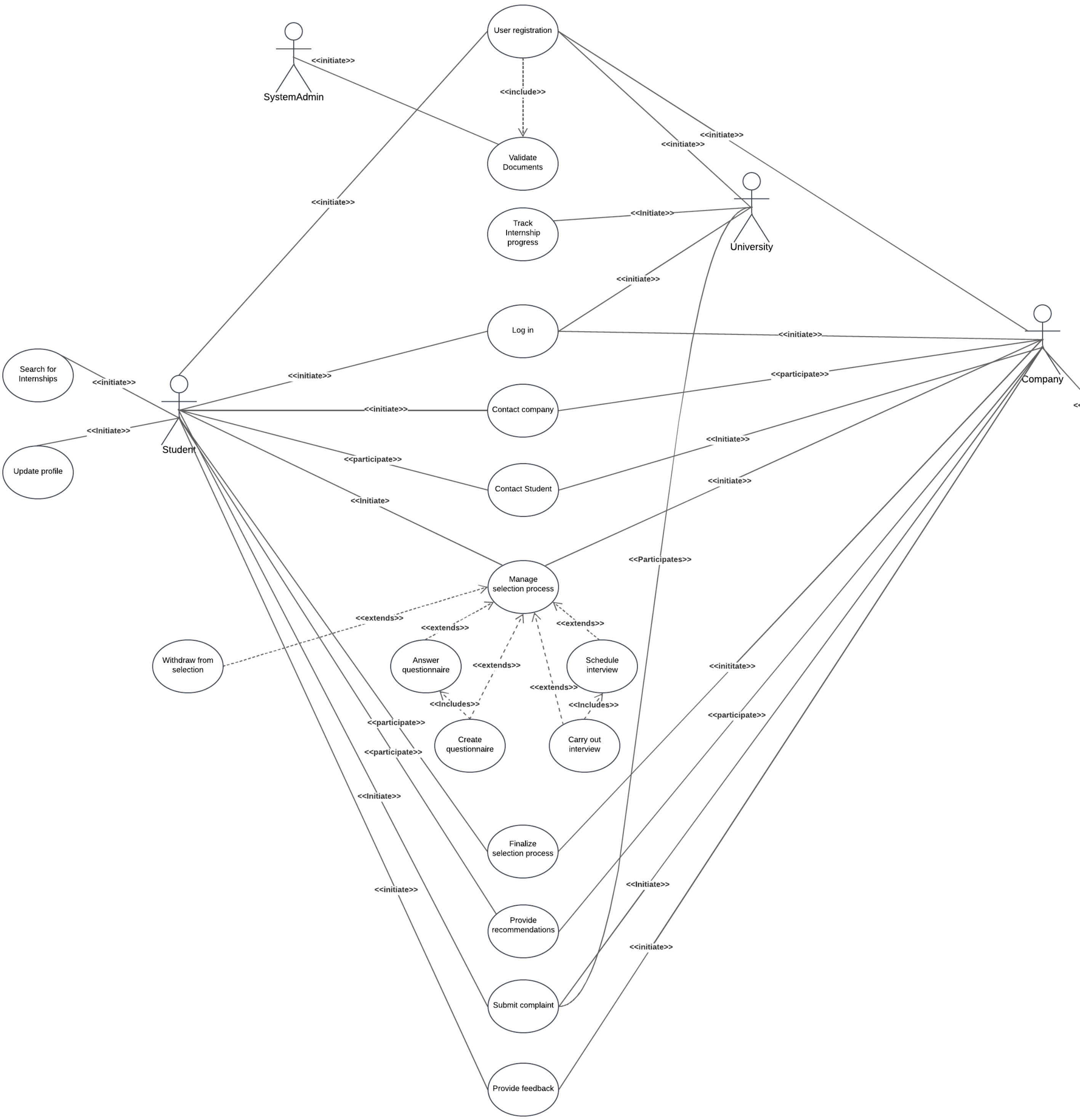

Figure 3.1: Use case Diagram

3.2.2. Use cases

UC1: User Registration

Actors	User (Student, company, or university)
Entry conditions	The user/institution is not already registered on the system and is on the main page.
Event flow	1. The user selects to sign-up as their corresponding role (sign-up as student, company, or university). 2. The user inputs their email and password. 3. The system sends a confirmation email. 4. The user confirms the email. 5. The system asks the user to provide personal data, CV, professional data, or institutional data, as well as official records to confirm the validity of the institution or the user's student status. 6. The user uploads the data requested. 7. The user is redirected to the main page.
Exit condition	The registration is successful, and the account is created, but it is still pending validation of the official documents to have access to all the functions of the system.
Exceptions	• The user doesn't fill out all the mandatory fields. The user is notified about the fields missing. • There is an account already linked to that email. The user is notified. • The user tries to upload their official proofs in an incorrect format. The user is notified about the format discrepancy. • The user does not fill data that is not mandatory during the registration. They can upload it later. • The user does not confirm the email. The registration procedure cannot continue.

Table 3.1: Use Case: User Registration

UC2: Validation of Documents

Actors	System administrator
Entry conditions	A user is registered and has sent their official documents for validation.
Event flow	1. The system notifies the system administrator that new documents are available for validation. 2. The system administrator opens the page of the documents. 3. The system administrator reviews the documents. 4. The system administrator determines whether the documents work as valid proof. 5. The system administrator updates the status of the account according to the final decision. 6. The user is notified about the update on their account.
Exit condition	The documents are valid, the account is validated, and its status is updated to "Approved," granting it access to the main functions.
Exceptions	• The system administrator decides the documents are not valid proof of the official status. The documents are rejected, and the status of the account is updated to "Rejected."

Table 3.2: Use Case: Validation of Documents

UC3: User Log-in

Actors	User (Student, company, university)
Entry conditions	The user has an account registered in the system and is on the main page.
Event flow	1. User presses the log-in button. 2. User inputs the email address and password linked to the account. 3. The system checks the credentials inputted. 4. The user is redirected to the main page.
Exit condition	The user has successfully logged in.
Exceptions	• User leaves a field empty. The user is notified. • Invalid email or password. The user is notified.

Table 3.3: Use Case: User Log-in

UC4: Search for Internships

Actors	Student
Entry conditions	The student is logged in, their account's status must be "Approved," and they are on the main page.
Event flow	1. The student selects the search bar. 2. The student filters the internship offers by keywords, range of salary, or different options. 3. The system looks for internships that match the filters. 4. The system returns to the student the matching internships.
Exit condition	The student comes back to the main page.
Exceptions	• The student tries an invalid input in a filter field.

Table 3.4: Use Case: Search for Internships

UC5: Student Updates Profile

Actors	Student
Entry conditions	The student is logged in and their account's status must be "Approved."
Event flow	1. The student clicks on the button "Profile." 2. The system displays the personal data that was inputted by the student, as well as the missing fields that they can fill. 3. The student selects "Update data." 4. The system displays the form to input and update the data. 5. The student updates or uploads the data and saves changes.
Exit condition	The data is successfully stored, and the page of the student's profile is reloaded.
Exceptions	• The student inputs invalid formatted data. The student is notified. • The changes are not saved by election of the student; the data remains the same.

Table 3.5: Use Case: Student Updates Profile

UC6: Post Internship

Actors	Company
Entry conditions	The company is logged in and the status of their account is "Approved."
Event flow	1. The user clicks on the button "Post internship." 2. The system loads the form with the data fields to post an internship. 3. The user inputs the data related to the internship in the form, including but not limited to title, description, salary, expected skills, and profile. 4. The user clicks on the post internship button. 5. The user is redirected to the internship post page.
Exit condition	The internship offer is posted and stored correctly.
Exceptions	• The data the user tried to input or upload is not in a valid format. The user is notified.

Table 3.6: Use Case: Post Internship

UC7: Contact Company

Actors	Student, Company
Entry conditions	The student is logged in, the status of their account is "Approved," has uploaded their CV, and is on the main page.
Event flow	1. The student clicks on an internship offer post that they find interesting. 2. The student clicks on the "Start contact" button to apply. 3. The system notifies the company about a new request. 4. The company accepts the contact request. 5. The selection process starts. 6. The system notifies the student and the company.
Exit condition	The selection process starts correctly after both parties accept the contact.
Exceptions	• The company is not interested in the contact request from the student, so they reject or ignore it, and the process does not start.

Table 3.7: Use Case: Contact Company

UC8: Contact Student

Actors	Company, Student
Entry conditions	The company is logged in, the status of their account is "Approved," and is on the main page.
Event flow	1. The company clicks on a student profile they believe might suit their needs. 2. The company clicks on the "Initiate contact" button to request a contact. 3. The system notifies the student about a new contact request. 4. The student accepts the contact request. 5. The selection process starts. 6. The system notifies the student and the company.
Exit condition	The selection process starts correctly after both parties accept the contact.
Exceptions	• The student is not interested in the contact request from the company, so they reject or ignore it, and the process does not start.

Table 3.8: Use Case: Contact Student

UC9: Manage Selection Process

Actors	Company, Student
Entry conditions	The user is logged in, the status of their account is "Approved," and is on the main page.
Event flow	1. The user clicks on the button "Selection processes" to see the selection processes they're involved in. 2. The system loads all the selection processes linked to the user. 3. The user selects the selection process they want to manage. 4. The user can select the options to manage the process or add events according to their role. 5. The other party is notified about the change or the new event in the selection process. 6. The selection process page is reloaded.
Exit condition	The change in the selection process is saved, and both parties are notified about it.
Exceptions	• There is an error when adding a new event. The user is notified.

Table 3.9: Use Case: Manage Selection Process

UC10: Create Questionnaire

Actors	Company, Student
Entry conditions	The company is on the page of the selection process they want to manage.
Event flow	1. The company clicks on "Post new questionnaire." 2. The system loads a form to create the questionnaire. 3. The company inputs all the questions. 4. The company personalizes the questionnaire by deciding which questions are open fields or multiple options. 5. The company inputs the correct answers. 6. The company saves and posts the questionnaire. 7. The company is redirected to the page of the selection process.
Exit condition	The questionnaire is saved and posted correctly on the selection process page, and the student linked is notified.
Exceptions	• The company tries to input questions or answers in invalid formats. • The company does not fill out answers to open field questions. They are notified after the questionnaire is answered to review the answers and grade them manually.

Table 3.10: Use Case: Create Questionnaire

UC11: Answer Questionnaire

Actors	Student, Company
Entry conditions	The student is on the page of the selection process they want to manage, and there is a questionnaire available to be answered.
Event flow	1. The student clicks on the now available button "Answer questionnaire." 2. The system loads the questionnaire. 3. The student chooses their answers and fills out the open field questions. 4. The student saves their answers and sends the questionnaire. 5. The system stores the answers. 6. The student is shown their grade.
Exit condition	The answers are saved and stored, the student is given their final grade, and the company is notified about the completion of the questionnaire.
Exceptions	• The student tries to input invalid data in a text field. The student is notified. • The company did not set answers to all open field questions. They are notified when the questionnaire is answered and are asked to review and grade the answers. The student is redirected to the selection process page.

Table 3.11: Use Case: Answer Questionnaire

UC12: Schedule Interview

Actors	Company, Student
Entry conditions	The company is on the page of the selection process they want to manage.
Event flow	1. The company clicks on "Schedule interview." 2. The system loads the company's calendar. 3. The company selects the hour and duration of the interview. 4. The interview date and hour are saved in the calendar of the company. 5. The interview date and hour are saved in the student's calendar.
Exit condition	The interview schedule is correctly saved in both calendars, and the student is notified.
Exceptions	• The company tries to schedule the interview in an already busy time for the student, in which case they are notified and must select a new timeframe.

Table 3.12: Use Case: Schedule Interview

UC13: Carry Out Interview

Actors	Company, Student
Entry Conditions	The company and the student are on the page of the selection process and an interview is scheduled.
Event Flow	1. The system notifies both the student and the company when it's almost the interview time. 2. The interviewer in the company account and the student click on "Join interview." 3. The student and interviewer can join with audio and video as they wish. 4. The interviewer uses the templates to write down annotations and details about the student. 5. The interviewer finishes the interview, and the room closes. 6. The system saves the template and the annotations written by the interviewer.
Exit Condition	The interview finishes, and both parties are redirected to the page of the selection process. The annotations written by the interviewer are stored.
Exceptions	• The student or the interviewer does not join the interview call. They can send a message explaining the reason and propose a reschedule.

Table 3.13: Use Case: Carry Out Interview

UC14: Withdraw From Selection

<table>
<tr><td>Actors</td><td>Student, Company</td></tr>
<tr><td>Entry Conditions</td><td>The student is on the page of the selection process they want to manage.</td></tr>
<tr><td>Event Flow</td><td>1. The student clicks on “Withdraw from selection.”
2. The system asks the student if they want to continue with the request.
3. The student confirms.
4. The system asks the student to write a reason as to why they decided to withdraw.
5. The student inputs the reason and clicks on “Send.”
6. The system displays a success banner.
7. The student is redirected to the main page.
8. The company is notified about the withdrawal from the student.</td></tr>
<tr><td>Exit Condition</td><td>The student withdraws from the selection successfully, and the selection process ends, changing its status to “Withdrawn.”</td></tr>
<tr><td>Exceptions</td><td>• The student decides not to withdraw the application and does not continue with the process. The selection process page is shown again.</td></tr>
</table>

Table 3.14: Use Case: Withdraw From Selection

UC15: Finalize Selection Process

Actors	Company, Student
Entry Conditions	The company is on the page of the selection process they want to manage.
Event Flow	1. The company clicks on "Finalize selection process." 2. The system displays a form to fill out the completion of the selection process. 3. The company states whether the result is selected or rejected for the internship and can provide reasons for the decision. 4. The company clicks on save, and the process is finalized. 5. The student is notified about the finalization of the process and the result.
Exit Condition	The company selects the student, and the internship is set to start. The student and company now have access to an "Internship" page.
Exceptions	• The company rejects the student, and the selection process ends.

Table 3.15: Use Case: Finalize Selection Process

UC16: Provide Recommendations

Actors	Student, Company
Entry Conditions	The company and the student must be registered and have “Approved” status in their accounts. The company has at least one internship offer posted.
Event Flow	1. The system runs the keyword-matching function. 2. The function returns pairs of students and internship offers that match based on preferences and professional profiles. 3. The system runs the rating function on all recommendations for each student and company. 4. The function returns a ranked list of recommendations for each student or company based on feedback and ratings. 5. The system selects the top 3 recommendations and notifies each student and company.
Exit Condition	The recommendations are saved and sent to the students and companies.
Exceptions	• There is no recommendation match for any student or company. No recommendation is given until at least one match is found.

Table 3.16: Use Case: Provide Recommendations

UC17: Submit Complaint

Actors	Student, Company
Entry Conditions	The student is enrolled in an ongoing internship with the company. The university related to the student has an "Approved" registered account in the system. The user giving the complaint is on the internship page.
Event Flow	1. The user clicks on "Send complaint." 2. The system displays a banner asking the user to write down the reason for the complaint. 3. The user fills out the text with the complaint. 4. The user saves and submits the complaint. 5. The system displays a success banner and saves the complaint.
Exit Condition	The complaint is saved and forwarded to the university.
Exceptions	• The user decides not to send a complaint and cancels the process. The internship page is shown again.

Table 3.17: Use Case: Submit Complaint

UC18: Track Internship Progress

Actors	University
Entry Conditions	The university has an "Approved" registered account, and one of their students is doing an ongoing internship. The university is on the main page.
Event Flow	1. The university clicks on "My students." 2. The university selects the student they want to check. 3. The system displays the information regarding complaints given to and by the student, as well as the progress achieved during the internship. 4. The university reviews the state of the internship and decides internally what to do.
Exit Condition	The university goes back to the main page or the page of their students.
Exceptions	• The university does not have a student in an ongoing internship. The "My students" page is empty.

Table 3.18: Use Case: Track Internship Progress

UC19: Provide Feedback

Actors	Student, Company
Entry Conditions	The student is enrolled in an ongoing internship with the company.
Event Flow	1. The company clicks on "End internship." 2. The system asks if the user is sure. 3. The company confirms. 4. The system displays a banner asking for feedback, suggestions, and a rating for the experience of the company with the student. 5. The company fills the form and sends the feedback. 6. The feedback from the company is saved and stored, and the system displays a success banner. 7. Once the student logs onto the platform, they will be asked to leave feedback about their experience with the company.
Exit Condition	Feedback from the company is saved and stored. When the student logs onto the platform and fills the feedback form, it will also be saved and stored.
Exceptions	• The company or student tries to close the page without leaving feedback. When they log in again, they will be asked to leave feedback for their experience.

Table 3.19: Use Case: Provide Feedback

3.2.3. Sequence Diagrams

UC1: User Registration

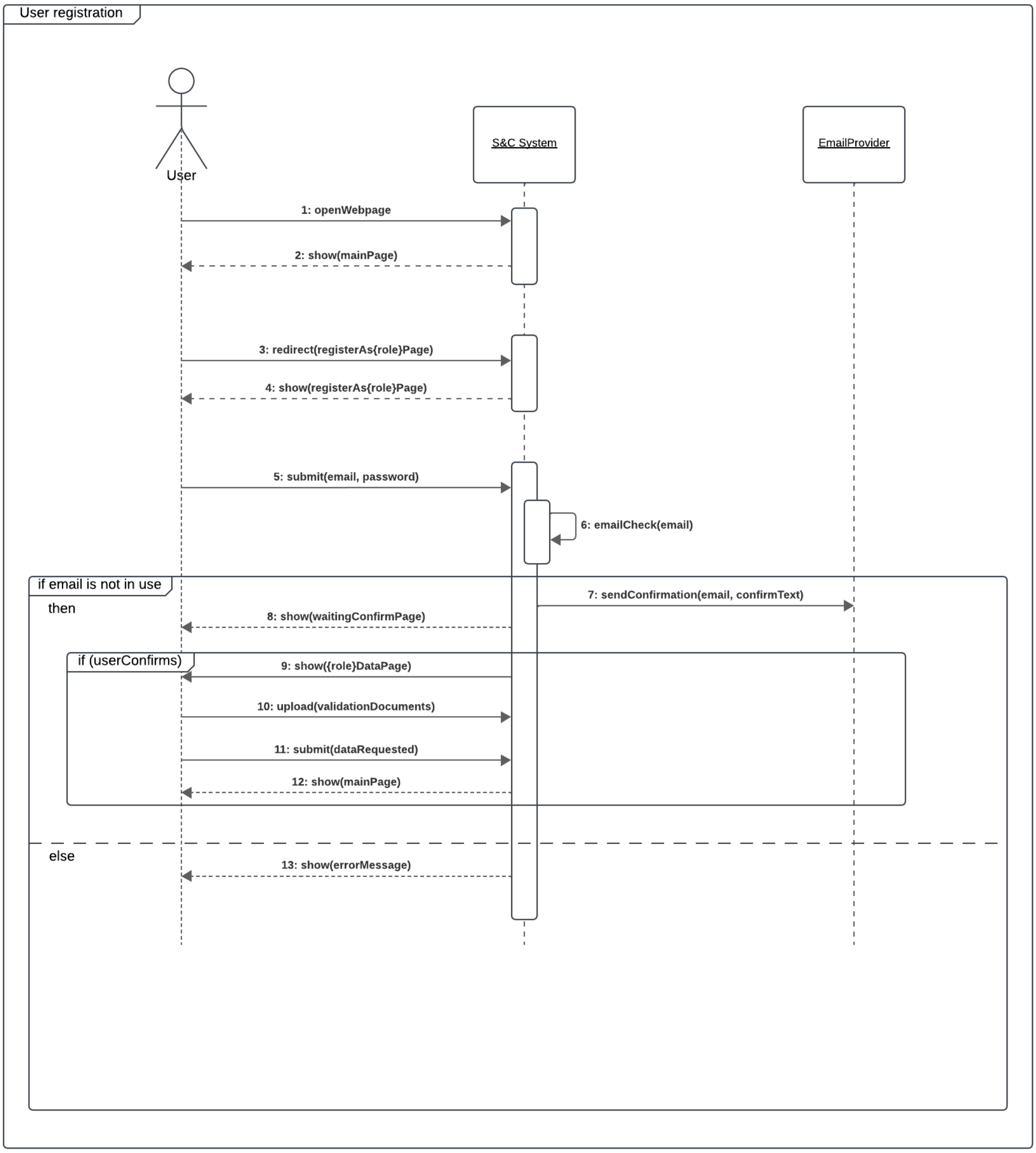

Figure 3.2: Sequence Diagram for UC1: User Registration

UC2: Validation of Documents

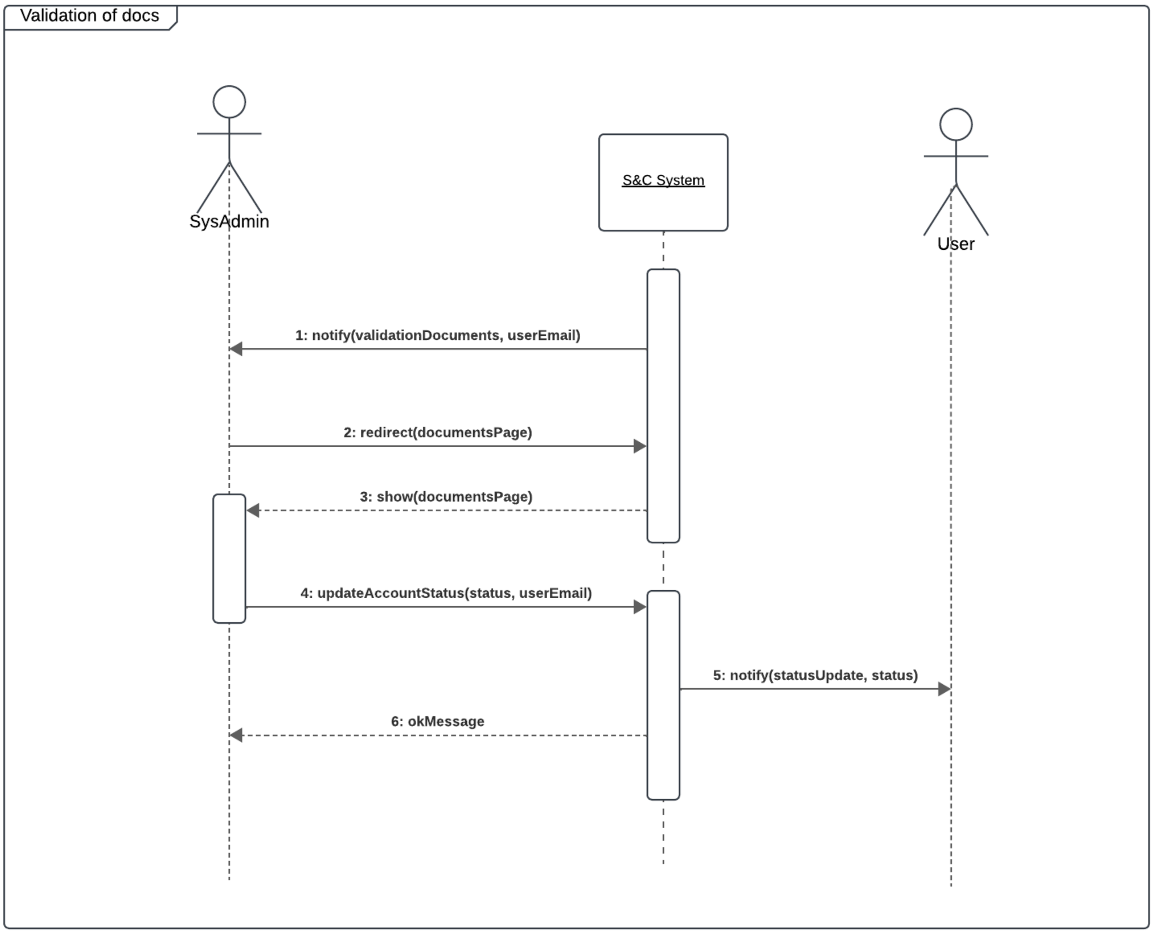

Figure 3.3: Sequence Diagram for UC2: Validation of Documents

UC3: User Log-in

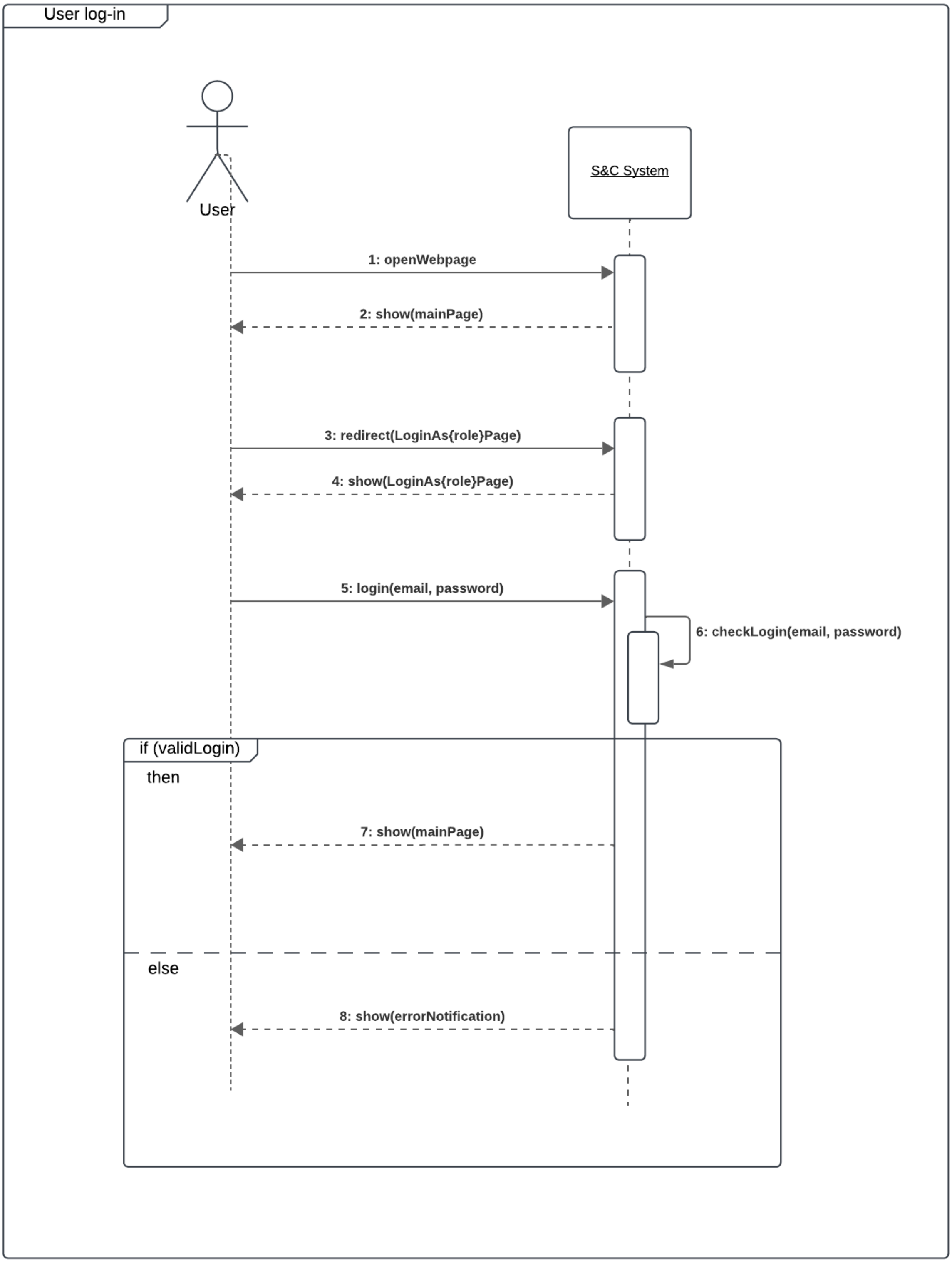

Figure 3.4: Sequence Diagram for UC3: User Log-in

UC4: Search for Internships

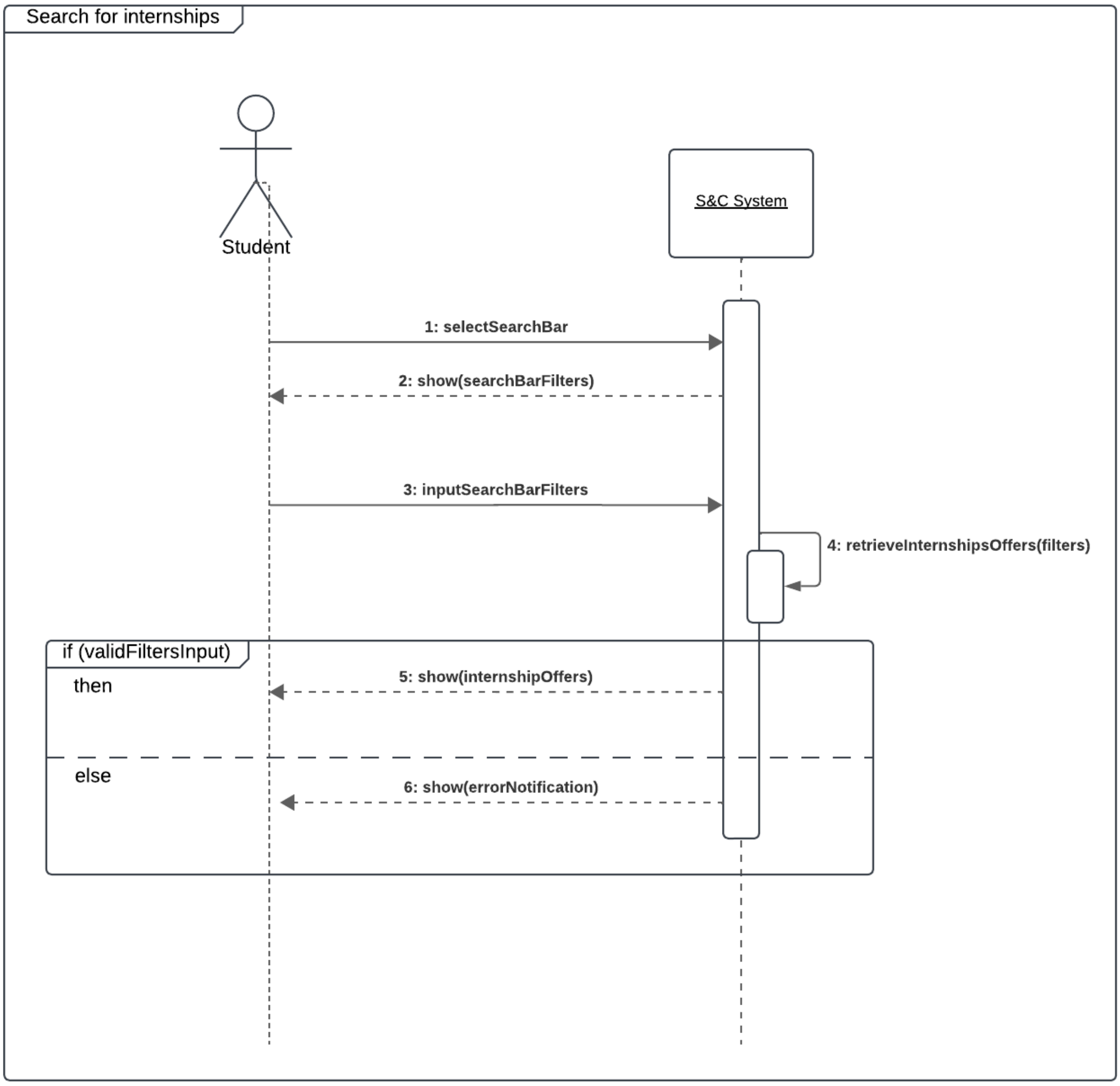

Figure 3.5: Sequence Diagram for UC4: Search for Internships

UC5: Student Updates Profile

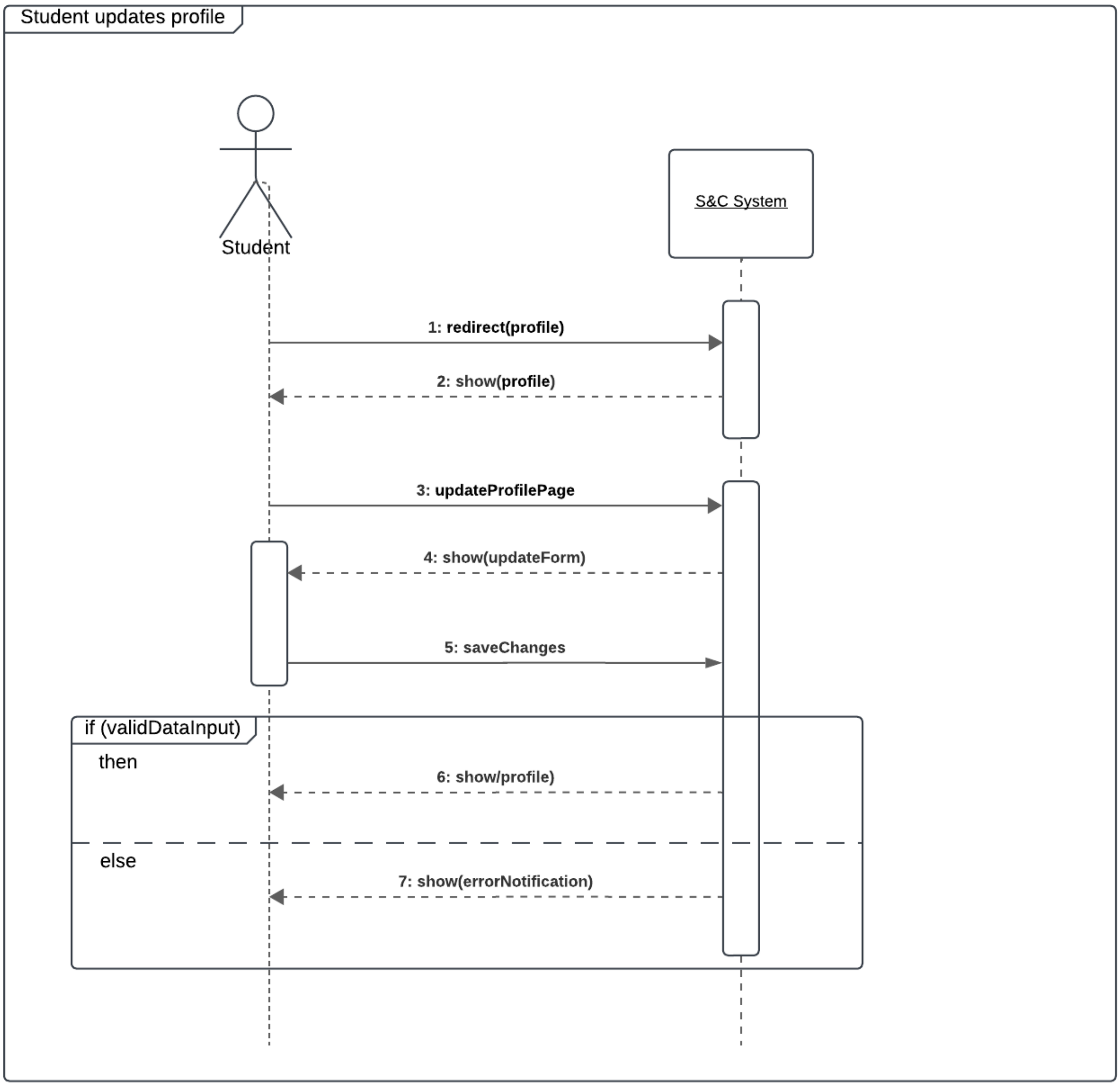

Figure 3.6: Sequence Diagram for UC5: Student Updates Profile

UC6: Post Internship

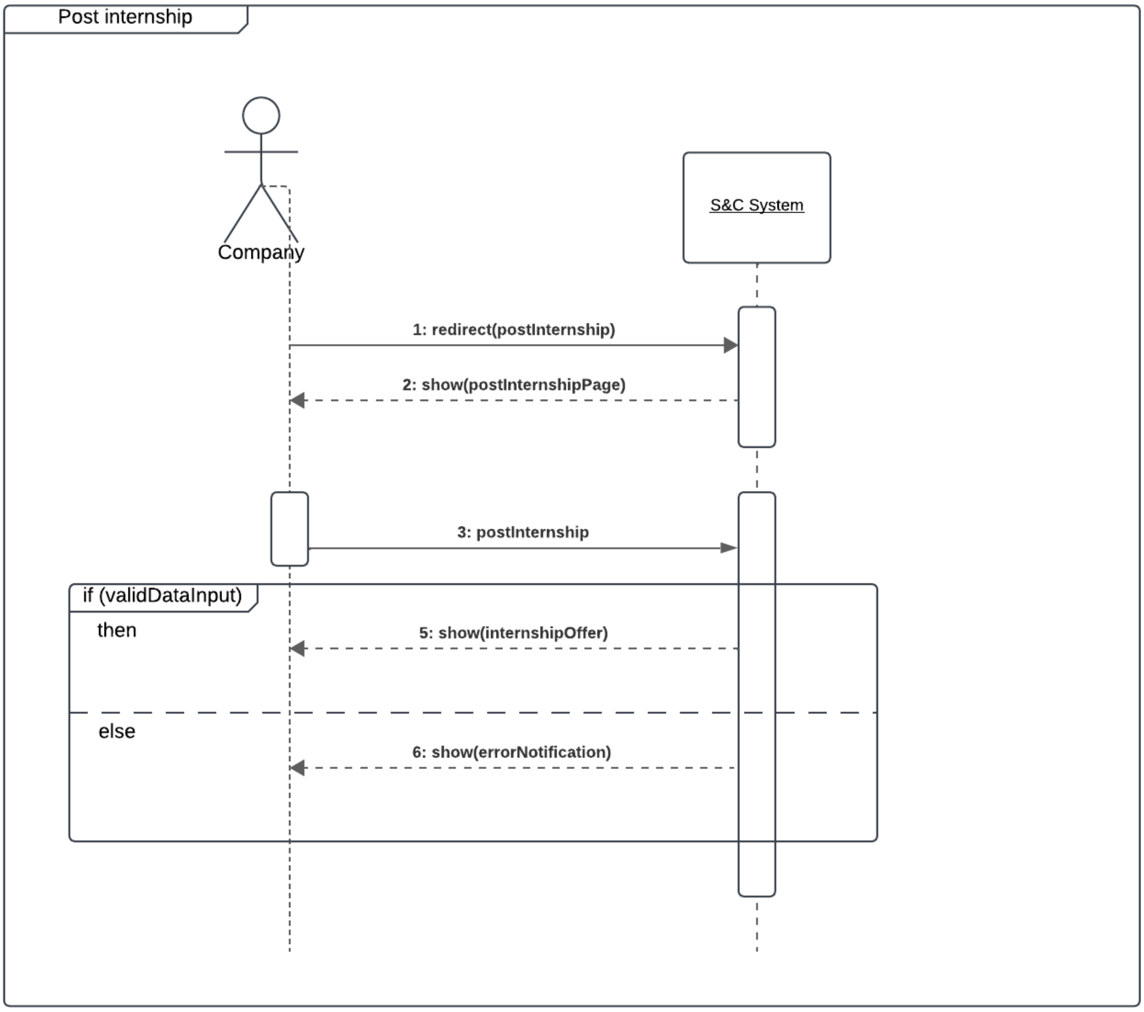

Figure 3.7: Sequence Diagram for UC6: Post Internship

UC7: Contact Company

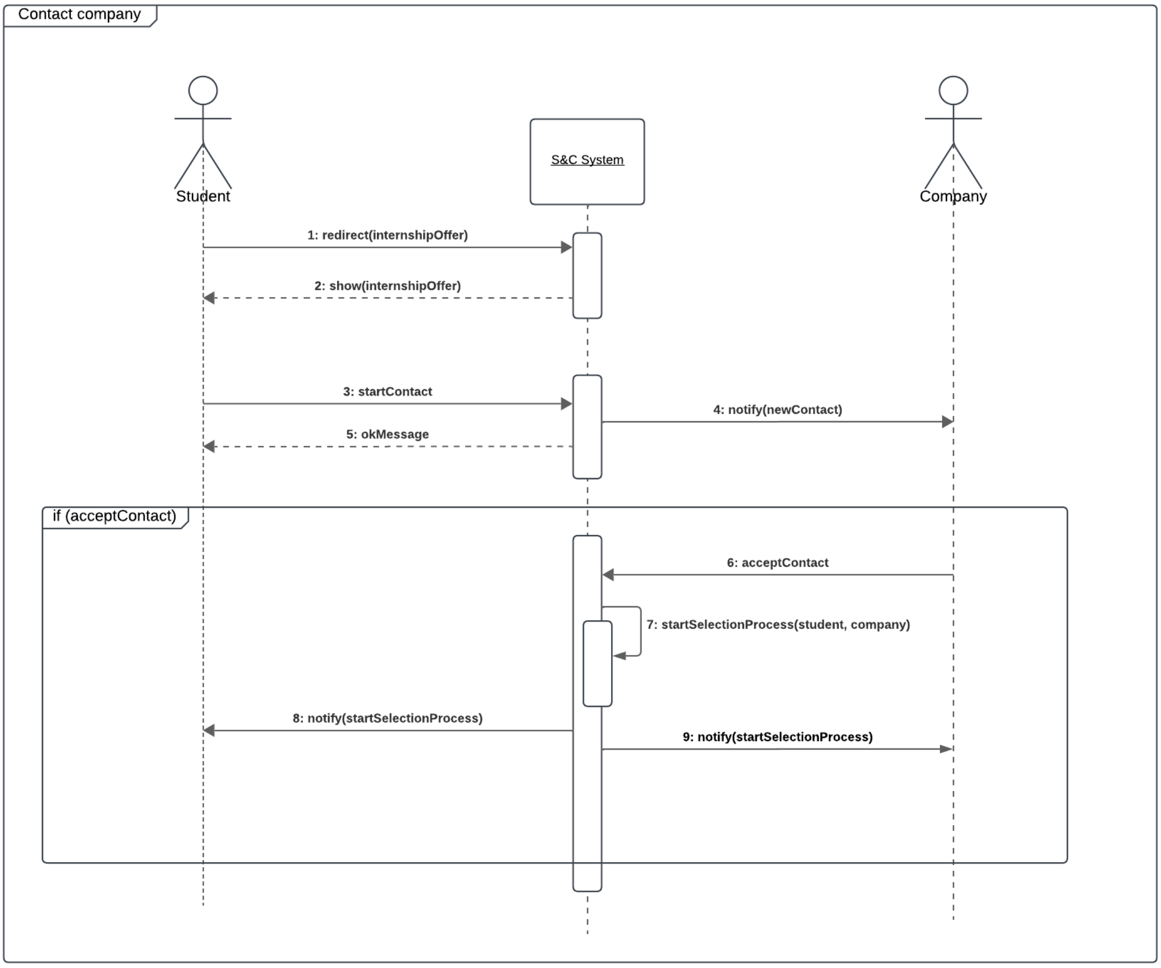

Figure 3.8: Sequence Diagram for UC7: Contact Company

UC8: Contact Student

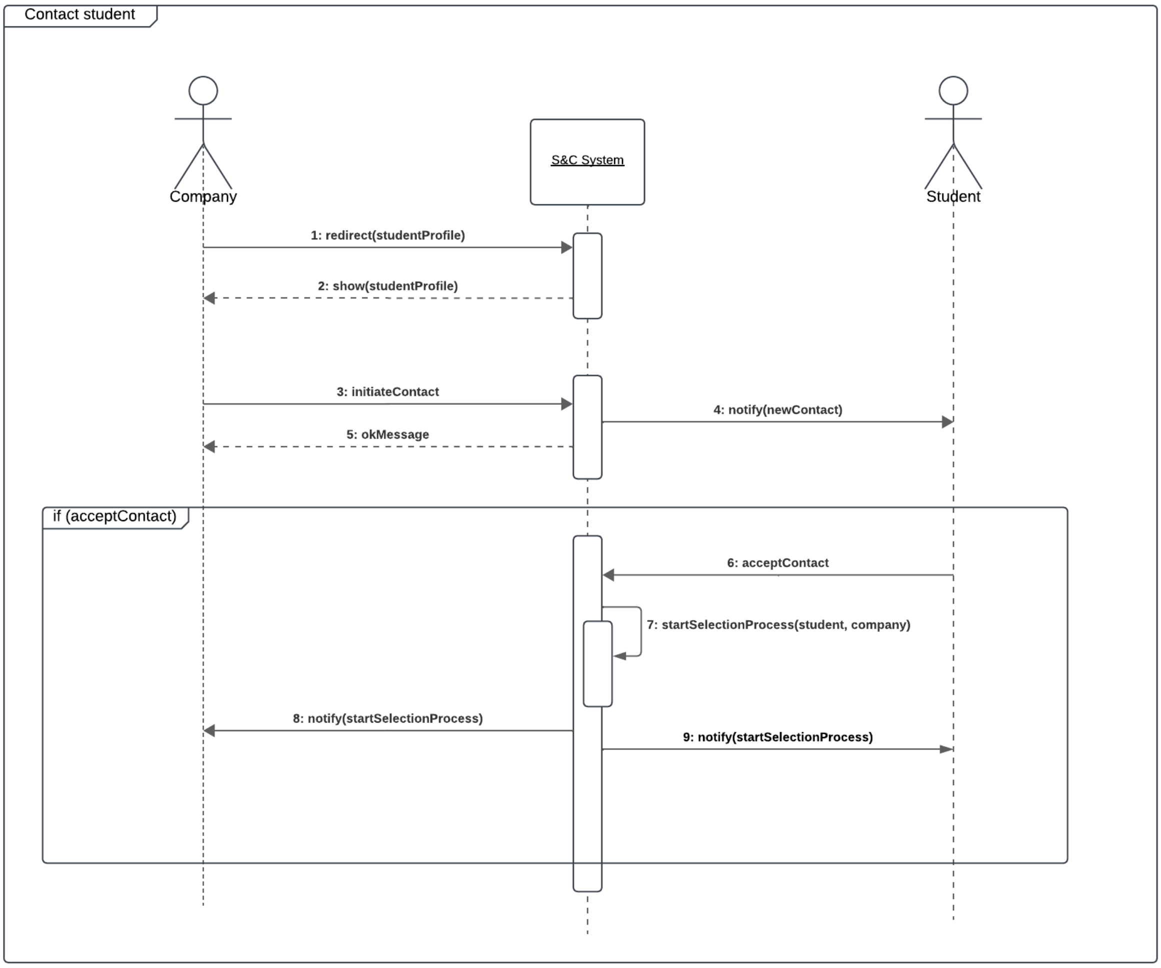

Figure 3.9: Sequence Diagram for UC8: Contact Student

UC9: Manage Selection Process

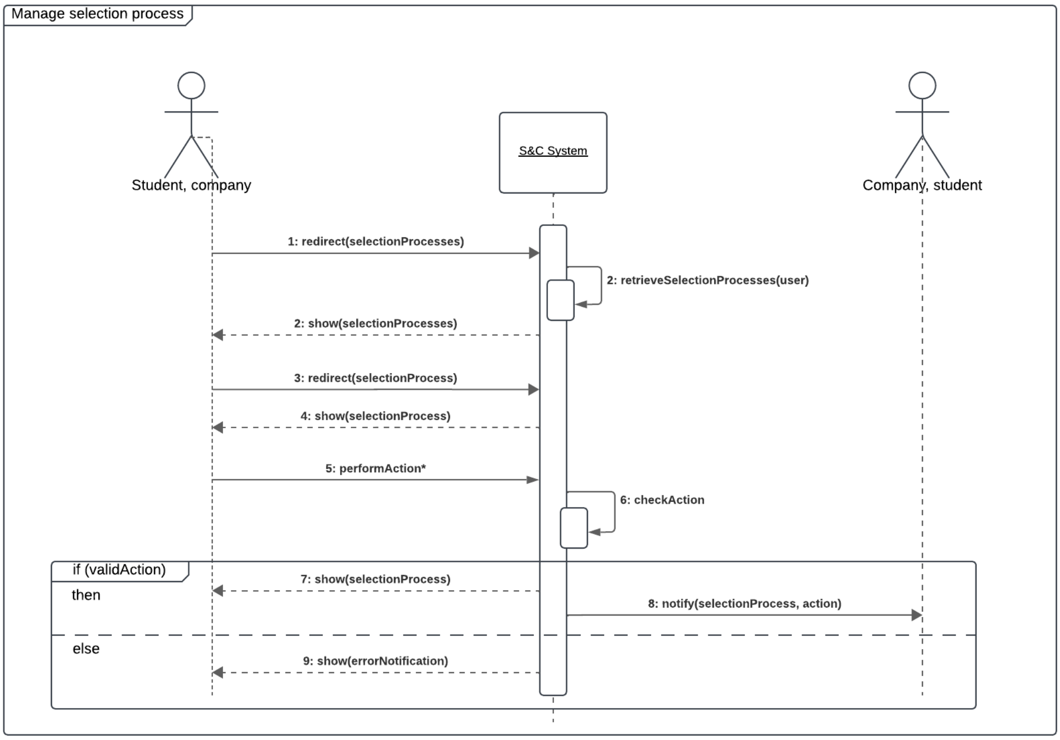

Figure 3.10: Sequence Diagram for UC9: Manage Selection Process

UC10: Create Questionnaire

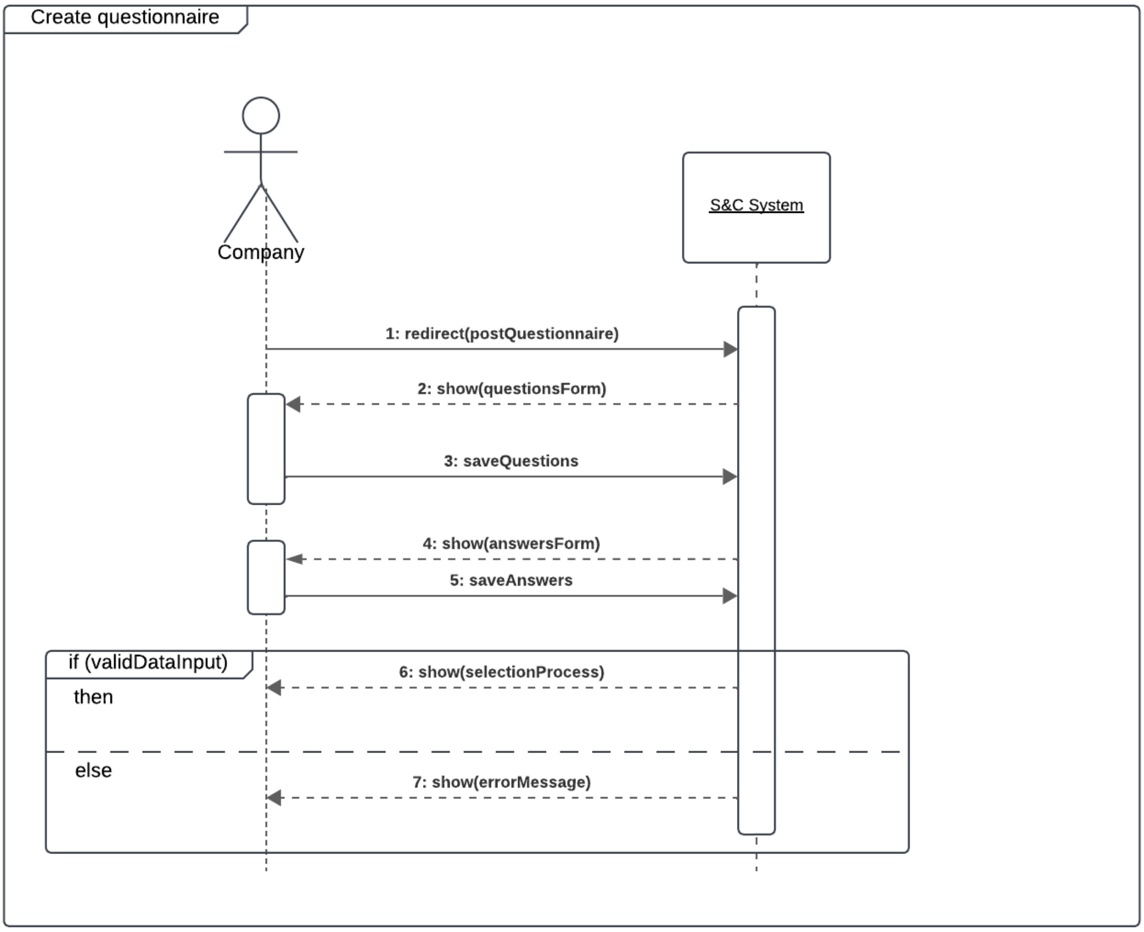

Figure 3.11: Sequence Diagram for UC10: Create Questionnaire

UC11: Answer Questionnaire

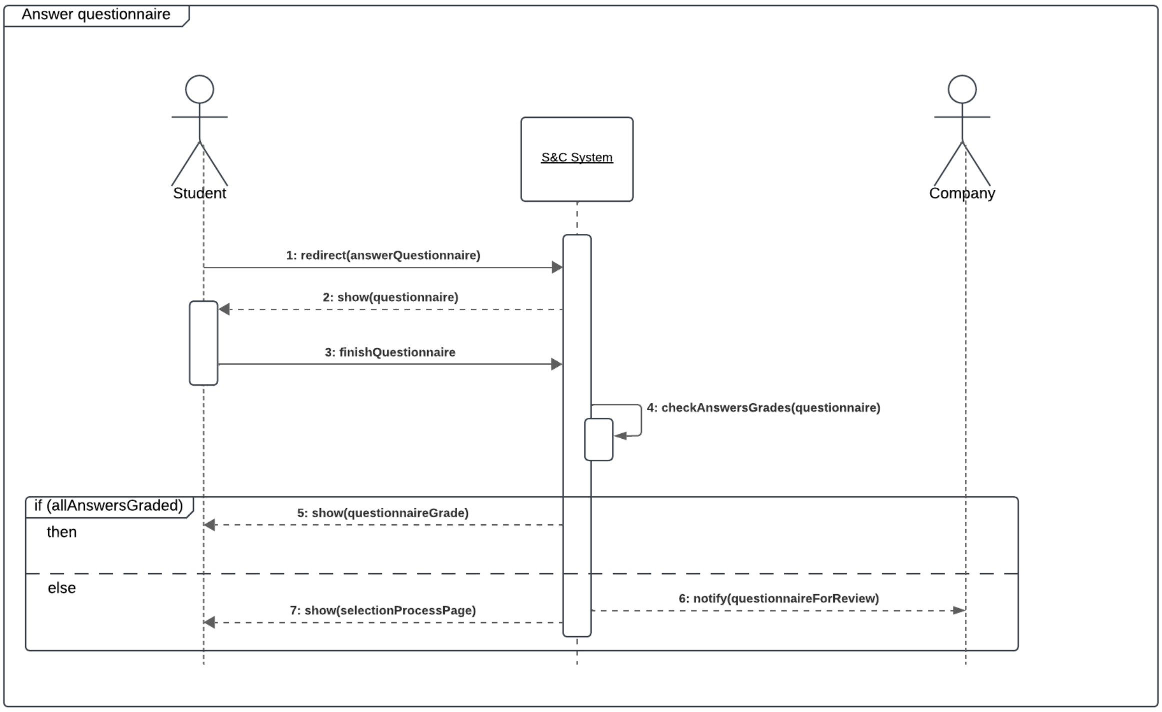

Figure 3.12: Sequence Diagram for UC11: Answer Questionnaire

UC12: Schedule Interview

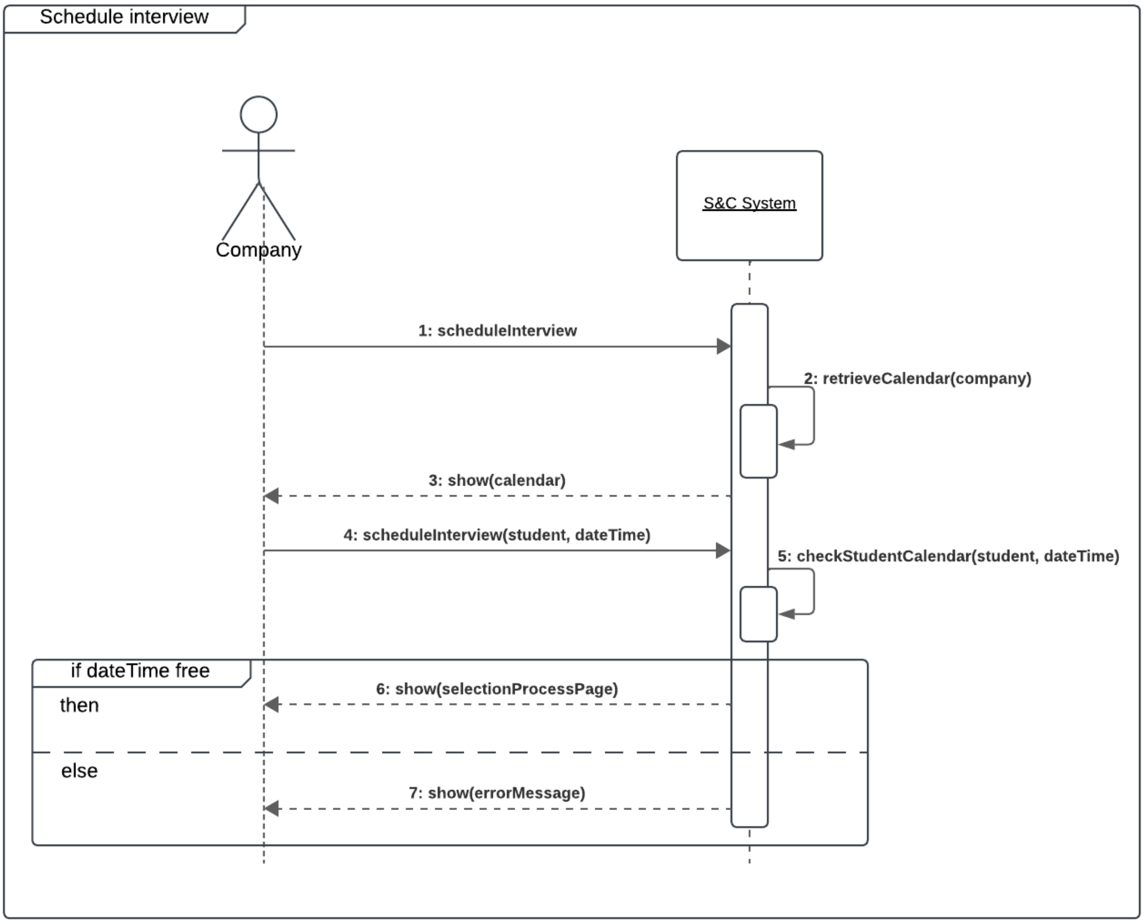

Figure 3.13: Sequence Diagram for UC12: Schedule Interview

UC13: Carry Out Interview

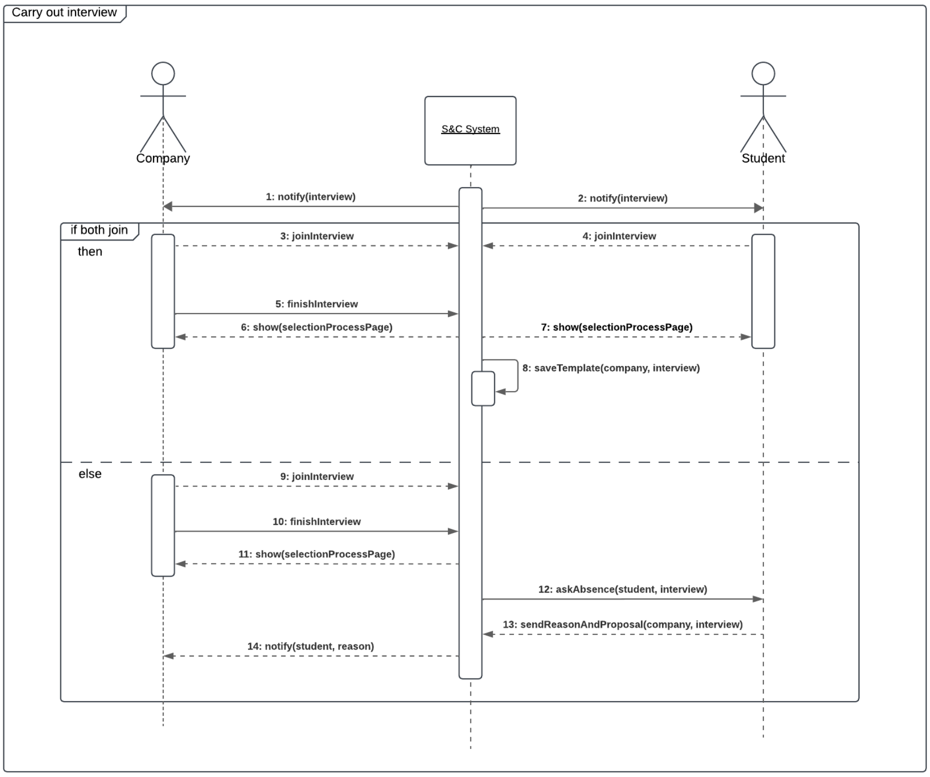

Figure 3.14: Sequence Diagram for UC13: Carry Out Interview

UC14: Withdraw from Selection

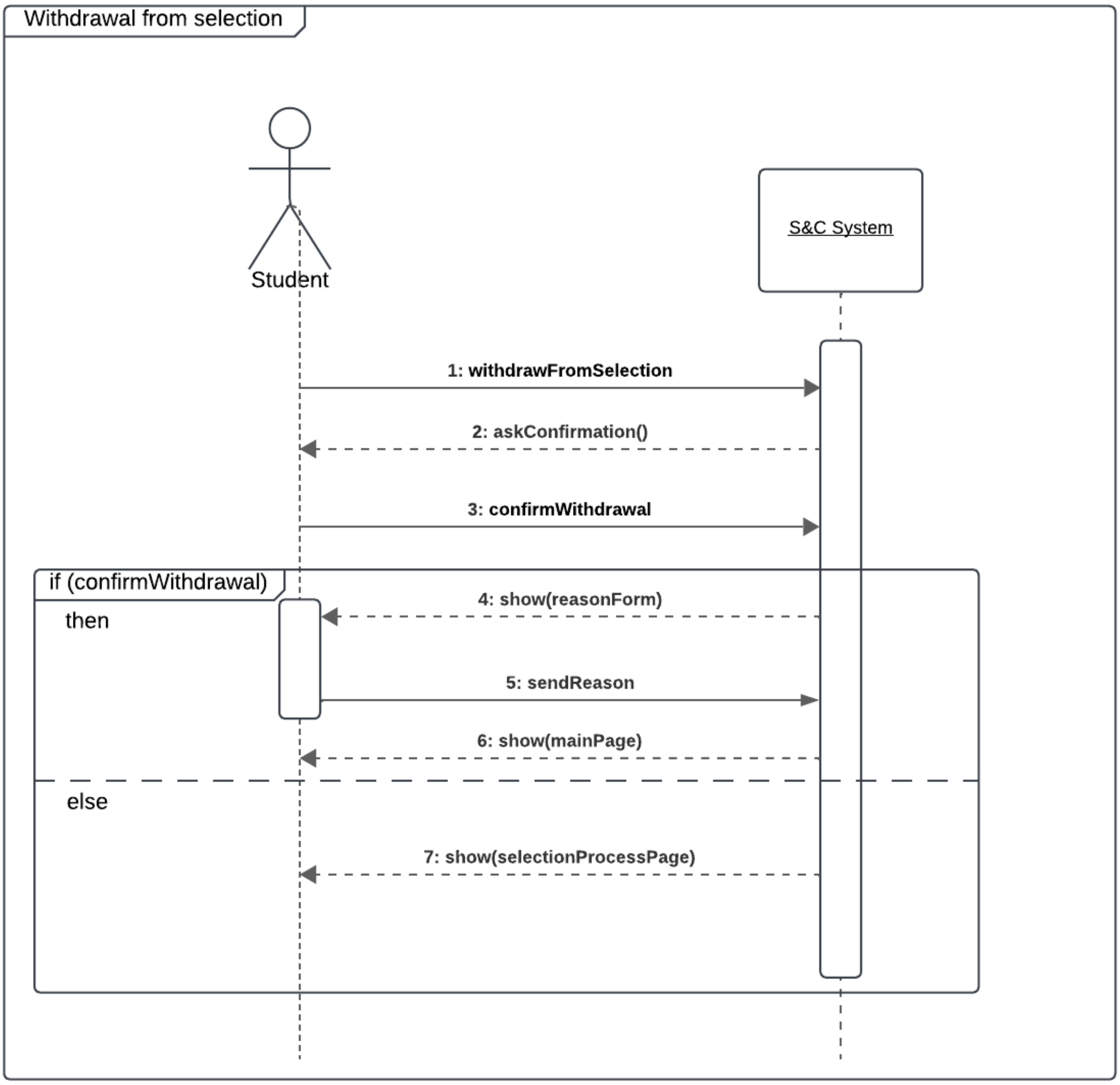

Figure 3.15: Sequence Diagram for UC14: Withdraw from Selection

UC15: Finalize Selection Process

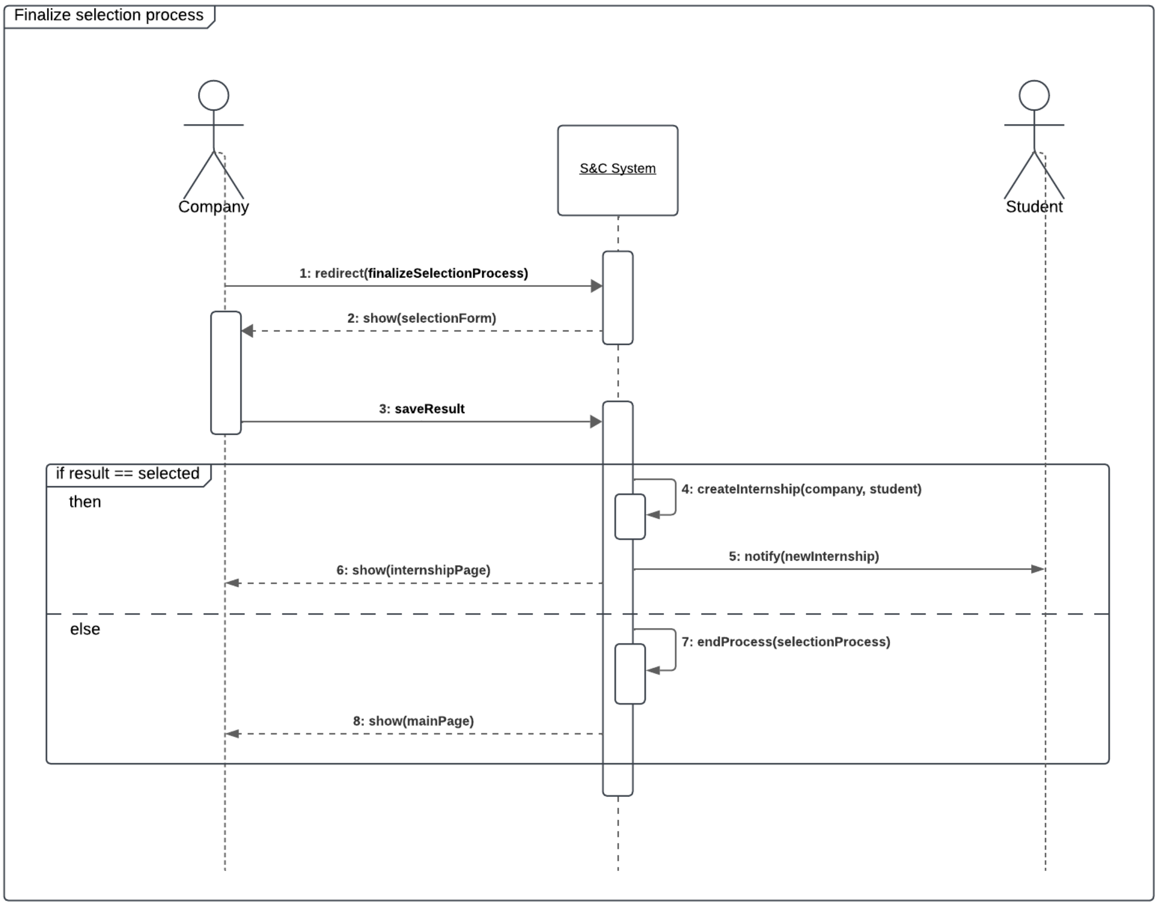

Figure 3.16: Sequence Diagram for UC15: Finalize Selection Process

UC16: Provide Recommendations

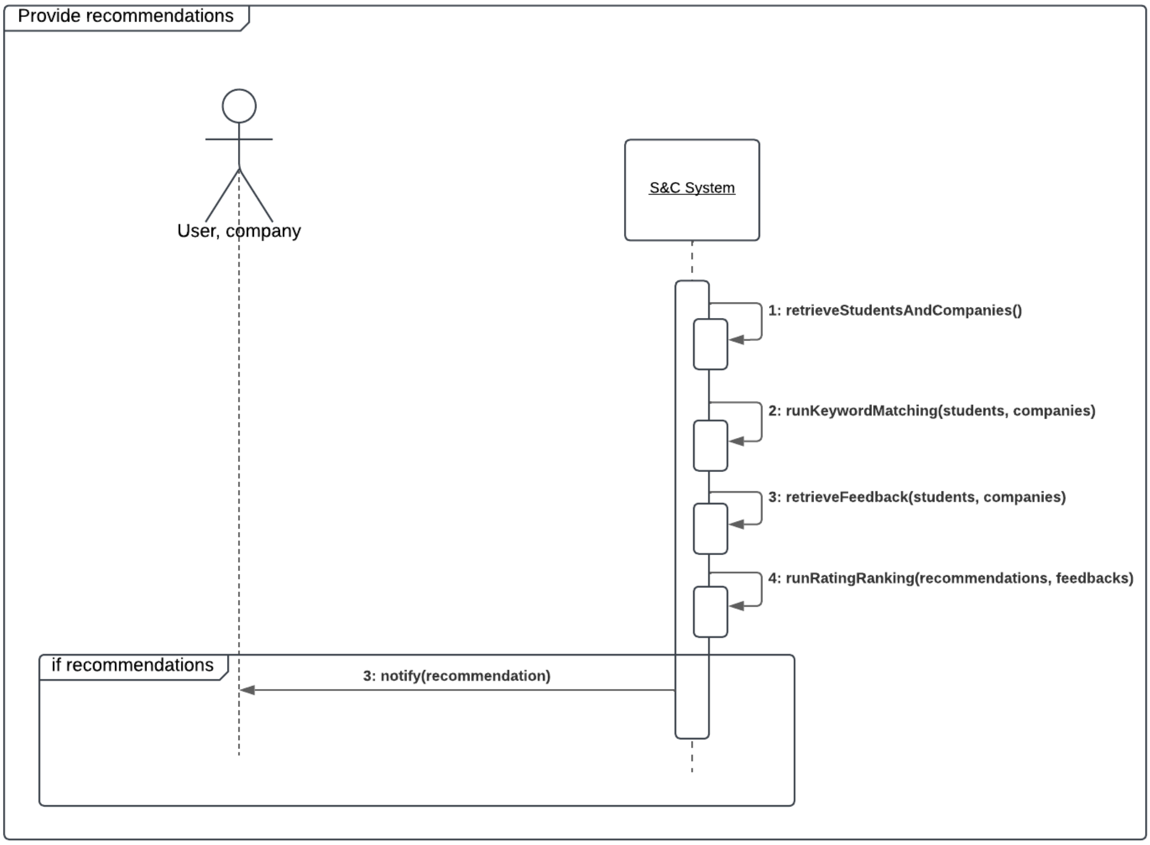

Figure 3.17: Sequence Diagram for UC16: Provide Recommendations

UC17: Submit Complaint

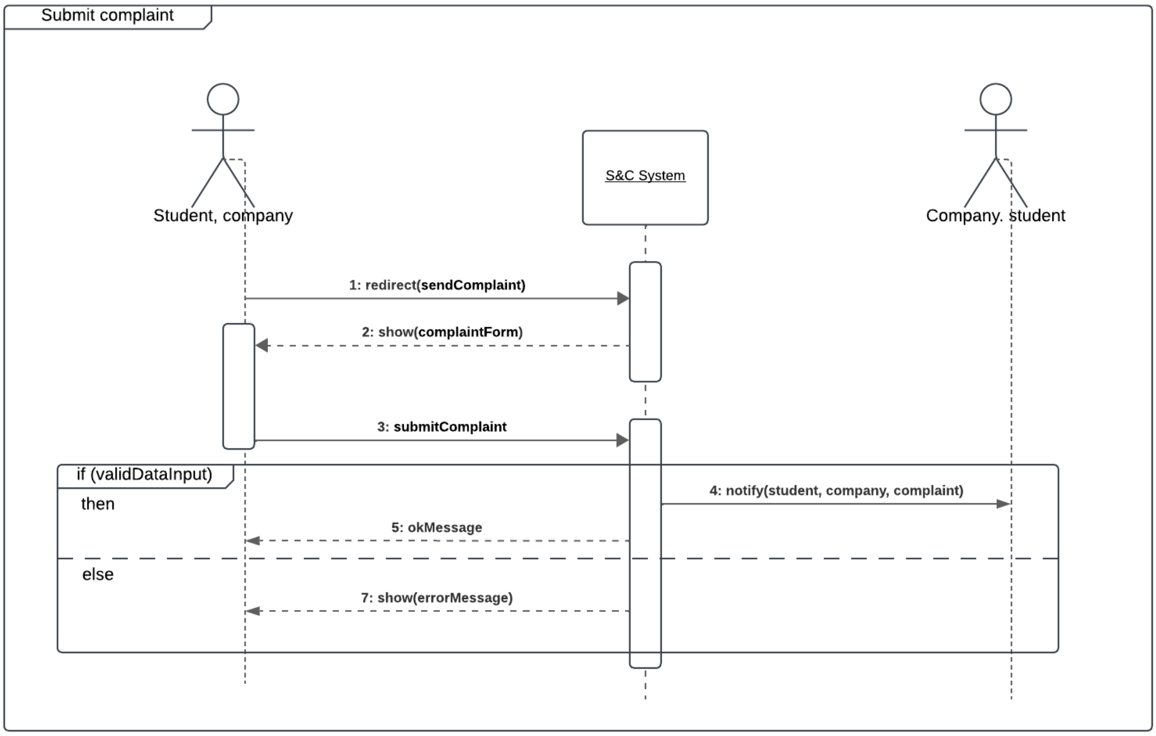

Figure 3.18: Sequence Diagram for UC17: Submit Complaint

UC18: Track Internship Progress

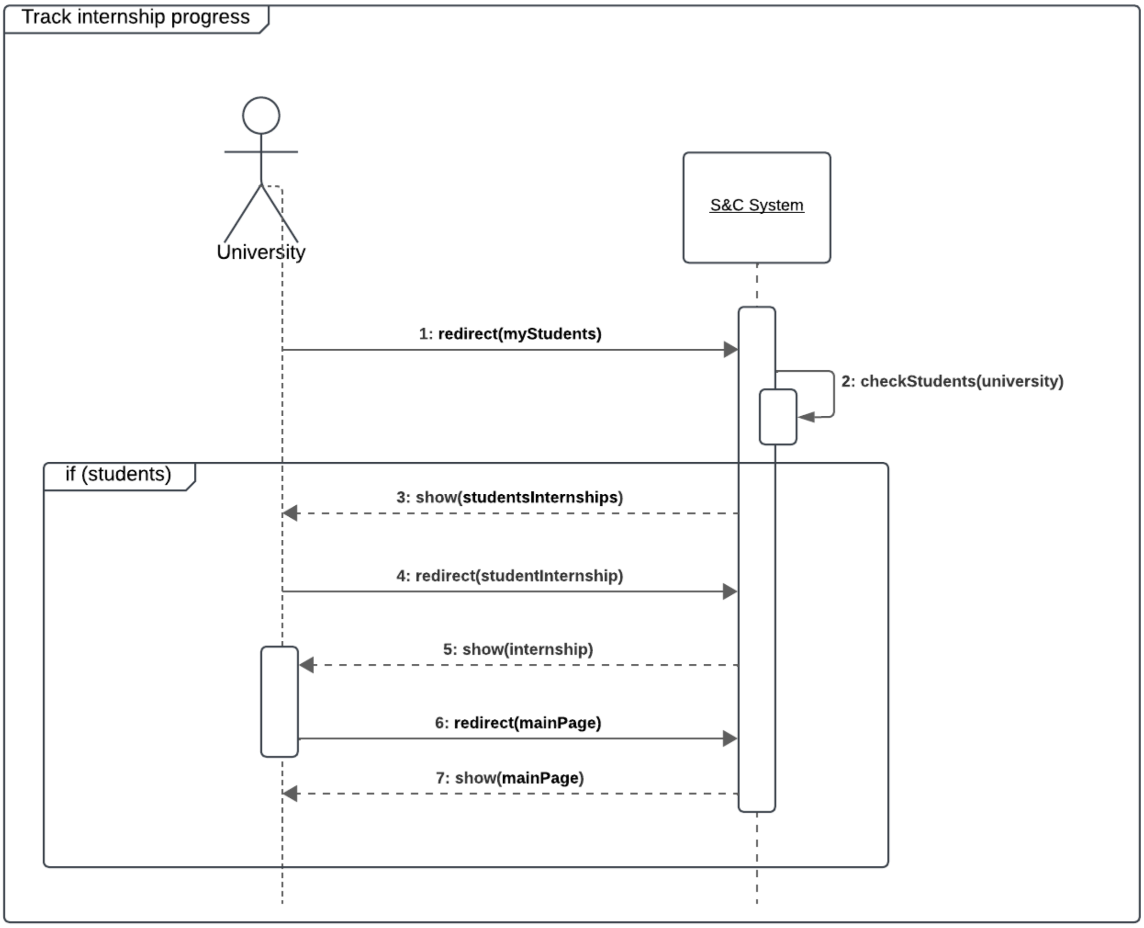

Figure 3.19: Sequence Diagram for UC18: Track Internship Progress

UC19: Provide Feedback

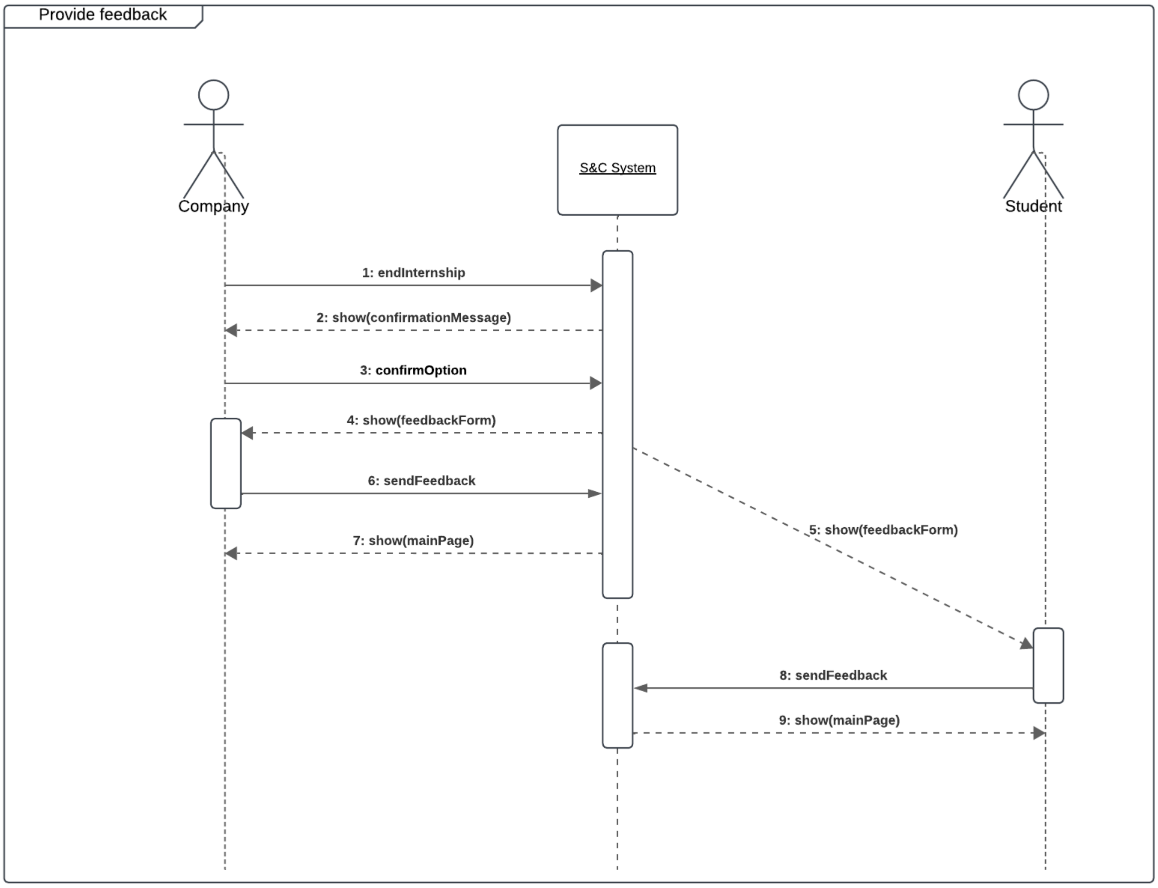

Figure 3.20: Sequence Diagram for UC19: Provide Feedback

3.2.4. Requirement mapping

Requirements Mapping for Goals

Goal G1: Students can find suitable internships that match their skills, experiences, and preferences.

Row ID	**Details**
G1	Students can find suitable internships that match their skills, experiences, and preferences.
Req ID	R1, R5, R8, R9, R12, R13, R15, R27, R6, R7, R17, R18, R19, R21, R26, R30
DA ID	D1, D2, D3, D4, D7
Use Case ID	UC1, UC2, UC3, UC4, UC5, UC7, UC9, UC11, UC13, UC14, UC16, UC17, UC19

Table 3.20: Requirements Mapping for Goal G1

Goal G2: Companies can find candidates that meet their internship requirements.

Row ID	Details
G2	Companies can find candidates that meet their internship requirements.
Req ID	R2, R5, R6, R7, R10, R11, R14, R16, R17, R19, R20, R21, R22, R23, R24, R25, R26
DA ID	D2, D5, D6, D7
Use Case ID	UC1, UC2, UC3, UC6, UC8, UC9, UC10, UC12, UC15, UC16, UC17, UC19

Table 3.21: Requirements Mapping for Goal G2

Goal G3: Students and companies can be recommended to one another based on their preferences, profile, and feedback.

Row ID	Details
G3	Students and companies can be recommended to one another based on their preferences, profile, and feedback.
Req ID	R8, R9, R10, R12, R13, R14, R31
DA ID	D3, D4, D5, D7
Use Case ID	UC4, UC5, UC6, UC16, UC19

Table 3.22: Requirements Mapping for Goal G3

Goal G4: Students and companies can monitor and manage the selection processes they're involved in.

Row ID	Details
G4	Students and companies can monitor and manage the selection processes they're involved in.
Req ID	R18, R19, R20, R21, R22, R23, R24, R25, R26, R27, R28
DA ID	D6
Use Case ID	UC9, UC10, UC11, UC12, UC14, UC15

Table 3.23: Requirements Mapping for Goal G4

Goal G5: Students, companies, and universities can monitor and track the status of internships, complaints, and the progress of candidates.

Row ID	Details
G5	Students, companies, and universities can monitor and track the status of internships, complaints, and the progress of candidates.
Req ID	R25, R27, R28, R29, R30, R31
DA ID	D1, D6, D7
Use Case ID	UC14, UC15, UC17, UC18, UC19

Table 3.24: Requirements Mapping for Goal G5

3.3. Non-Functional Requirements

3.3.1. Performance Requirements

The S&C platform must ensure high performance and responsiveness to provide an optimal user experience. Specific requirements include:

- **Responsiveness:** The delay between user requests and system responses should not exceed 1 second, assuming the user has a stable internet connection.
- **High Traffic Handling:** The system must handle peak loads efficiently, especially during high-demand periods, such as transitions between academic years, where user influx is expected to increase.

3.3.2. Design Constraints

Standards Compliance

To ensure reliability and compliance with industry standards, the platform must adhere to:

- **GDPR:** The system must handle private data securely and comply with the General Data Protection Regulation (GDPR) to ensure data protection and privacy for EU users.
- **Date and Time Standards:** S&C must implement the ISO 8601 standard for date and time formats, ensuring global usability.
- **Accessibility Standards:** The platform must comply with the Web Content Accessibility Guidelines (WCAG), enabling access for users with disabilities.

Hardware Limitations

Users are expected to access the platform using devices that meet the following minimum requirements:

- **Web Browsers:** The device should support modern web browsers and operate them seamlessly.
- **Internet Connectivity:** The platform requires a stable internet connection (2G/3G/4G/4.5G/5G/V Fi).

3.3.3. Software System Attributes

Reliability

Reliability is essential for the continuous functioning of the platform. Requirements include:

- **Concurrency Handling:** The system must support multiple concurrent users without errors in processing actions like initiating contact, recommendations, or interviews.

- **Data Integrity:** Backup systems and data recovery mechanisms must be in place to prevent data loss during failures.

Availability

Availability is critical to meet user expectations. Specific requirements are:

- **Uptime Targets:** The platform must achieve at least 99.9% availability.
- **Failover Mechanisms:** The system should include failover mechanisms to prevent downtime, avoiding single points of failure.

Security

Security measures must protect sensitive data and prevent unauthorized access. Key requirements include:

- **Data Encryption:** Data transmission must be encrypted using HTTPS to ensure confidentiality.
- **Access Control:** Implement role-based access control to restrict user permissions to relevant features and data.
- **Database Security:** Apply strong security measures, such as storing hashed passwords using robust cryptographic algorithms, to protect the database.

Maintainability

Maintainability ensures the platform can be updated and scaled efficiently. Requirements include:

- **Modular Design:** The system should be modular, with clean and well-documented code to support seamless updates and feature additions.
- **Monitoring Tools:** Real-time monitoring tools should be implemented to identify performance issues and improve the platform continuously.

Portability

Portability ensures that users can access the platform from a variety of devices. Specific requirements include:

- **Cross-Platform Compatibility:** The platform must be accessible via desktop and mobile devices.
- **Browser Support:** The system must operate smoothly on multiple modern browsers.
- **Operating System Independence:** Users should be able to access the platform from various operating systems.

4 | Formal Analysis Using Alloy

This Alloy model represents a system that manages internships, focusing on the interactions between universities, students, companies, internship offers, selection processes, and internships. The primary entities (or signatures) include University, Student, Company, InternshipOffer, SelectionProcess, and Internship. The relationships between these entities are carefully defined to ensure consistency and logical transitions, such as linking accepted selection processes to internships.

Key constraints enforce the uniqueness of student-university relationships, ensure students can only contact valid offers, and regulate the flow of statuses in the selection process. The model also specifies conditions for transitioning from pending statuses and guarantees that rejected or accepted statuses remain stable. The predicates and facts together ensure the integrity of the system and simulate valid scenarios.

4.1. Alloy Model for Internship Management System

This Alloy model represents a system designed to manage internships, encapsulating the relationships and interactions between universities, students, companies, and internships. Below is the Alloy model description:

```
sig University {
    has: set Student
}

sig Student {
    has: one ProfessionalProfile,
    uploads: one CV,
    contacts: set InternshipOffer // Students can contact multiple
        internship offers
}

sig ProfessionalProfile {}
sig CV {}

sig Company {
    posts: set InternshipOffer // The internship offers posted by the
        company
}

sig InternshipOffer {
    involves: set SelectionProcess,  // Some internship offers involve
        one or more selection processes
```

```
    postedBy: one Company            // The company that posted this
        internship offer
}

sig Questionnaire {} {
    one sp: SelectionProcess | this = sp.usesQuestionnaire
}

sig Template {} {
    one sp: SelectionProcess | this = sp.usesTemplate
}

sig Interview {} {
    one sp: SelectionProcess | this = sp.usesInterview
}

sig SelectionProcess {
    participant: one Student,
    usesQuestionnaire: lone Questionnaire,
    usesInterview: lone Interview,
    usesTemplate: lone Template,
    var status: one Status
} {
    one io: InternshipOffer | this in io.involves and io in participant.
        contacts
}

sig Internship {
    offeredBy: one Company,
    isPreceded: one SelectionProcess
} {
    one io: InternshipOffer | isPreceded in io.involves and (offeredBy =
         io.postedBy)
    isPreceded.status = Accepted
}

abstract sig Status {}
one sig Accepted, Pending, Rejected extends Status {}

fact UniqueStudentUniversity {
    all s: Student | one u: University | s in u.has
}

fact OfferMadeByCompany {
    all c: Company | all io: c.posts | io.postedBy = c
}

fact oneProfessionalProfile {
    all p: ProfessionalProfile | one s: Student | s.has = p
}

fact oneCV {
    all cv: CV | one s: Student | s.uploads = cv
}

```

```
fact StudentsContactOffers {
    all s: Student | s.contacts in Company.posts
}

fact UniqueSelectionProcess {
    all sp: SelectionProcess | lone io: InternshipOffer | sp in io.involves
}

fact NoDuplicateParticipants {
    all io: InternshipOffer | all sp1, sp2: io.involves | sp1 != sp2 => sp1.participant != sp2.participant
}

fact PendingOrRejectedNotLinkedToInternship {
    all sp: SelectionProcess | sp.status in (Pending + Rejected) => no i: Internship | i.isPreceded = sp
}

fact RejectedRemainsRejected {
    all sp: SelectionProcess | always (sp.status = Rejected =>  always sp.status = Rejected)
}

fact AcceptedRemainAccepted {
    all sp: SelectionProcess | always (sp.status = Accepted =>  always sp.status = Accepted)
}

fact PendingTransitions {
    all sp: SelectionProcess | sp.status = Pending => eventually (sp.status = Accepted or sp.status = Rejected)
}

fact InternshipAppearsWithAccepted {
    all sp: SelectionProcess | sp.status = Accepted => some i: Internship | i.isPreceded = sp
}

fact EventuallyAccepted {
    all io: InternshipOffer | #io.involves >= 2 => some sp: io.involves | eventually sp.status = Accepted
}

pred show {
    some io: InternshipOffer | #io.involves >= 2 and one Internship
}

run show
```

Listing 4.1: Alloy Model for Internship Management System

4.1.1. Model Explanation

- **Entities:** The primary entities in this model include universities, students, companies, and their respective attributes.
- **Relations:** Each student is linked to a university and can contact internship offers posted by companies. Companies post internship offers, which involve one or more selection processes.
- **Selection Process:** A selection process links a student and an internship offer and transitions between statuses: *Pending*, *Accepted*, or *Rejected*.
- **Internships:** Internships are only created from selection processes with an *Accepted* status.

4.1.2. Key Constraints and Facts

- **Unique Relationships:** Each student belongs to a single university and has one professional profile and CV.
- **Valid Transitions:** Pending statuses eventually transition to *Accepted* or *Rejected*, and once a status is set, it remains stable.
- **Internship Creation:** Internships are only created for accepted selection processes.
- **Consistency:** No two selection processes share the same participant or involve duplicate internships.

4.1.3. Execution

The predicate `show` demonstrates a scenario where an internship offer has multiple selection processes, and at least one internship is created.

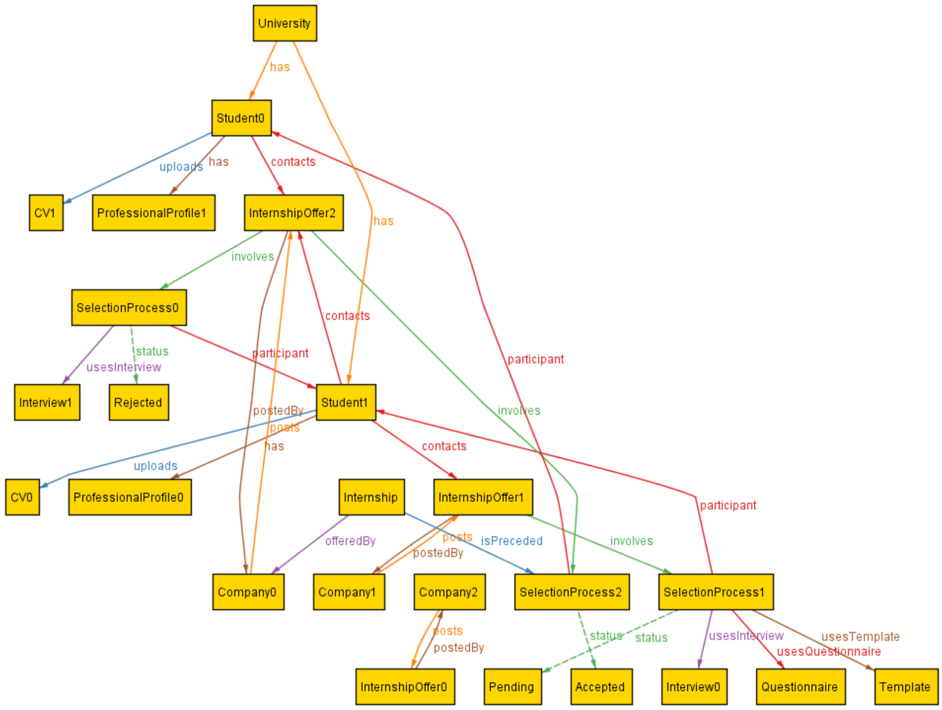

Figure 4.1: Alloy example pending status

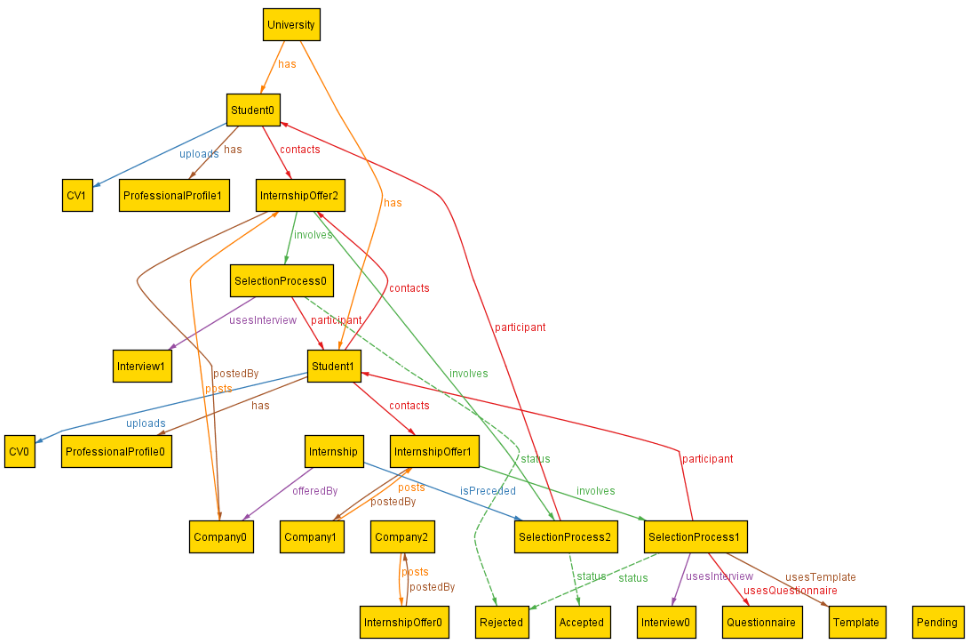

Figure 4.2: Alloy example Accepted/Rejected status

5 | Effort spent

5 Effort Spent

In this part, there is an overview of the time effort spent by each member of this team. Everyone has spent some time writing each section of this document, and here it is visible the amount of time.

- **Andrea Donado**

Chapter	Effort (In hours)
1	10
2	15
3	10
4	8

- **Miguel Sierra**

Chapter	Effort (In hours)
1	15
2	10
3	8
4	10

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